EML 4722 Computational Fluid Dynamics

CFD Modeling of Air Flow through a Subsonic Wind Tunnel using ANSYS Fluent

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Abstract

This project focuses on the development and optimization of a homemade, open-system wind tunnel designed and analyzed using ANSYS Fluent. The primary objective is to enhance the efficiency of the wind tunnel for aerodynamic testing. The base geometry of the wind tunnel follows standard subsonic dimensions, featuring an intake and diffuser aimed at stabilizing the flow and minimizing recirculation zones.

Three diffuser geometries and one inlet geometry were modeled and tested to assess their impact on flow characteristics within the wind tunnel. Utilizing ANSYS Fluent, simulations were conducted to analyze velocity distributions and recirculation zones. Results indicated that reducing the diffuser angle of attack from ±12.95° to ±10.20° significantly diminished recirculation, thereby improving the tunnel's flow uniformity and accuracy for experimental testing. Comparison of velocity vectors and pathlines across the diffuser designs demonstrated the effectiveness of a ±10° angle in minimizing disturbances post-test section. While a comparison including a tangential inlet design demonstrates an effective measure to reduce skin friction drag along the walls of the tunnel as well as a smaller but more uniform system of recirculation zones post-test section.

Uniformity testing across the tunnel's test section revealed consistent velocity profiles at key section lines, confirming the reliability of the optimized design. Future enhancements could explore extending the diffuser length or implementing a negative attack angle post-test section to further reduce recirculation zones and enhance pressure recovery.

In conclusion, optimizing the diffuser angle of attack is crucial for minimizing flow disturbances and improving the reliability of aerodynamic experiments conducted in homemade wind tunnels. This study underscores the importance of systematic design iterations and computational validation in achieving optimal performance of aerodynamic testing devices.

Table of Contents

1. Introduction

The design and analysis of wind tunnels have been a crucial aspect of aerodynamic research and development. This project aims to develop a homemade, open-system wind tunnel, modeled and tested in ANSYS Fluent. The primary objective is to design the most efficient wind tunnel that can test the flow around a small object. However, the focus is on optimizing the tunnel itself rather than analyzing the object inside. The inclusion of an object is optional and secondary to the primary goal of enhancing the tunnel performance.

The study of airflow within wind tunnels is vital to understanding and predicting the behavior of various aerodynamic applications. An efficient wind tunnel provides stable, uniform flow conditions, which are essential for obtaining accurate and reliable results in experiments. This project addresses the need for an efficient, homemade wind tunnel that can be used for educational purposes and small-scale aerodynamic testing.

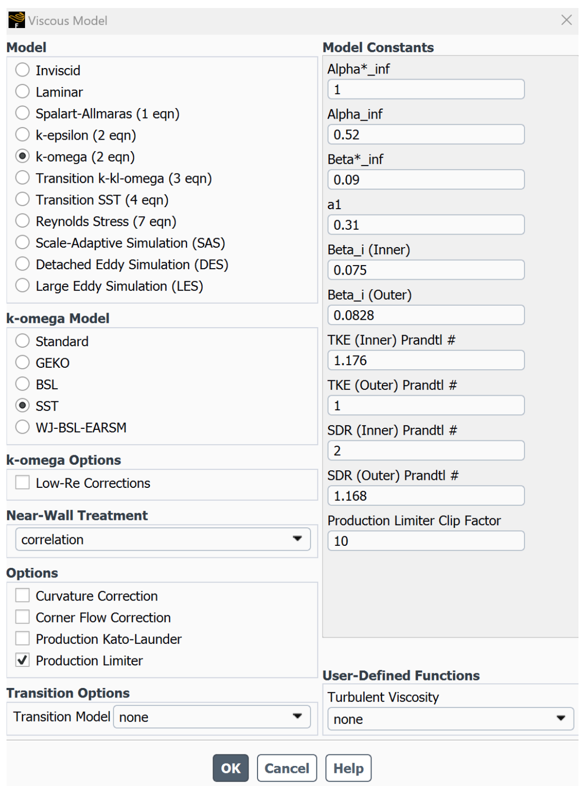

Figure 0-1: Viscous Model

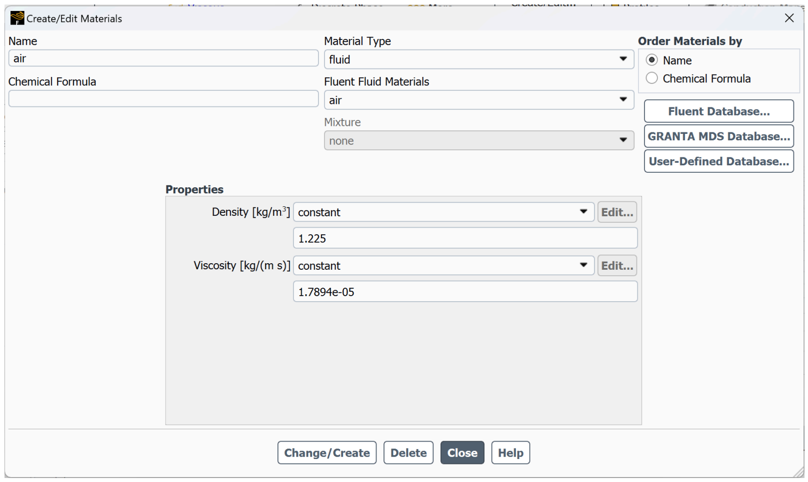

Figure 0-2: Material Properties

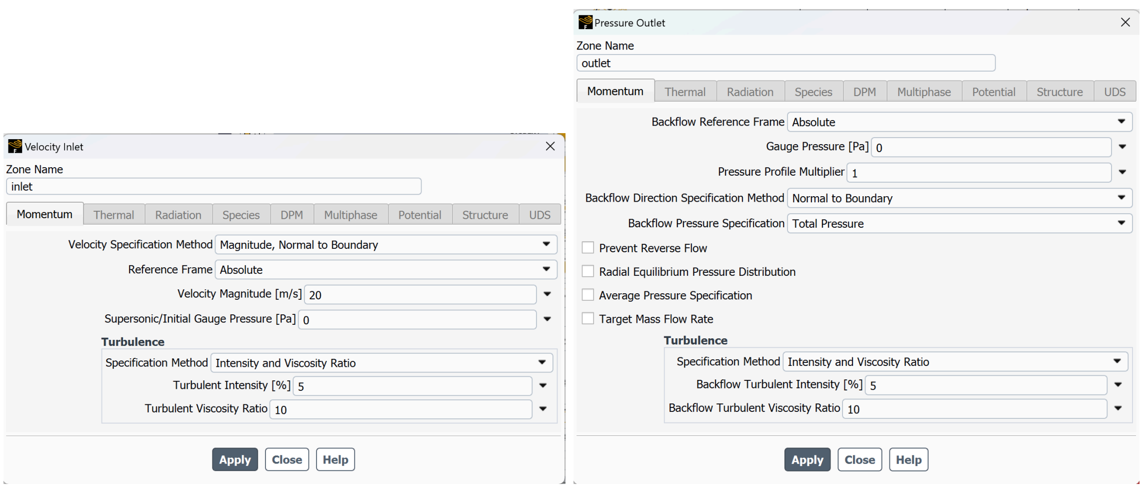

Figure 0-3: Boundary Conditions

Base Geometry Dimensions

The base geometry is built based on standard dimensions of a subsonic wind tunnel. The extension of framing of the intake is a 60x60mm opening with a height of 25mm. The intake is a truncated square pyramid with a lower base of 60x60mm, a height of 50mm, and an upper base of 26x26mm. The intake leads into the 26x26mm test area that extends 27mm towards the diffuser. The diffuser is a truncated square pyramid with an upper base that connects to the test area of 26x26mm, a height of 100mm, and an outlet lower base of 46x46mm. See Figure 3-1 for the geometry of the base wind tunnel tested.

Modifications will be made to the diffuser prior to further testing as we believe this will reduce the recirculation zone, creating a more accurate test area and a more consistent testing device.

Diffuser

The diffuser decelerates the high-speed flow from the test section, achieving static pressure recovery and reducing the load on the drive system. The flow field within the diffuser is influenced by the characteristics of the flow exiting the test area. Factors such as orientation, size, and wake development in the air flow affect the diffuser entrance flow. To prevent flow separation, the diffuser's area should increase gradually along the axis. [3]

The diffuser geometry will use a 10:1 scale of the base design for modeling and ease of use purposes. Diffuser geometry version 1, our base, starts from a square inlet that is 60x60mm with a diffuser that extends 100mm from the test area to a square that is 46x46mm. The diffuser geometry version 2, starts from a square inlet that is 60x60mm with a diffuser that extends 100mm from the test area to a square that is 40x40mm. The diffuser geometry version 3, starts from a square inlet that is 60x60mm with a diffuser that extends 100mm from the test area to a square that is 36x36mm.

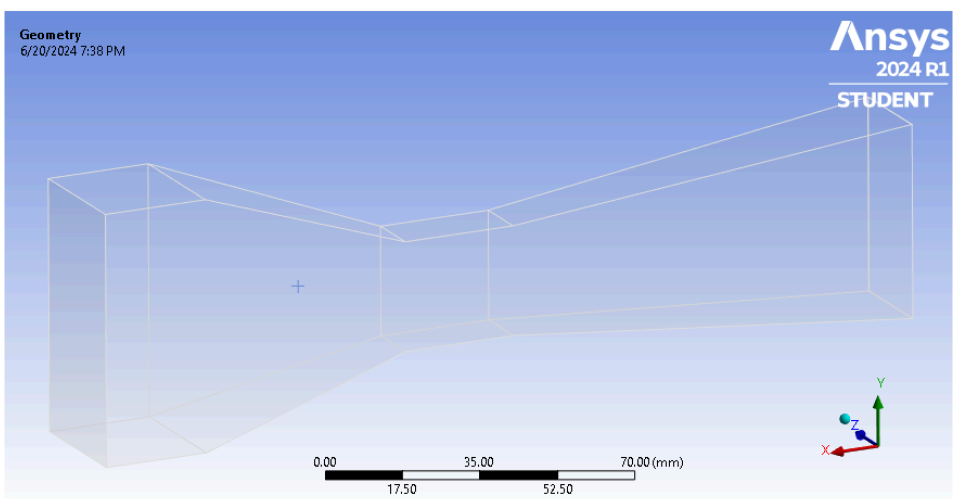

Figure 3-1: Diffuser 46x46mm Geometry

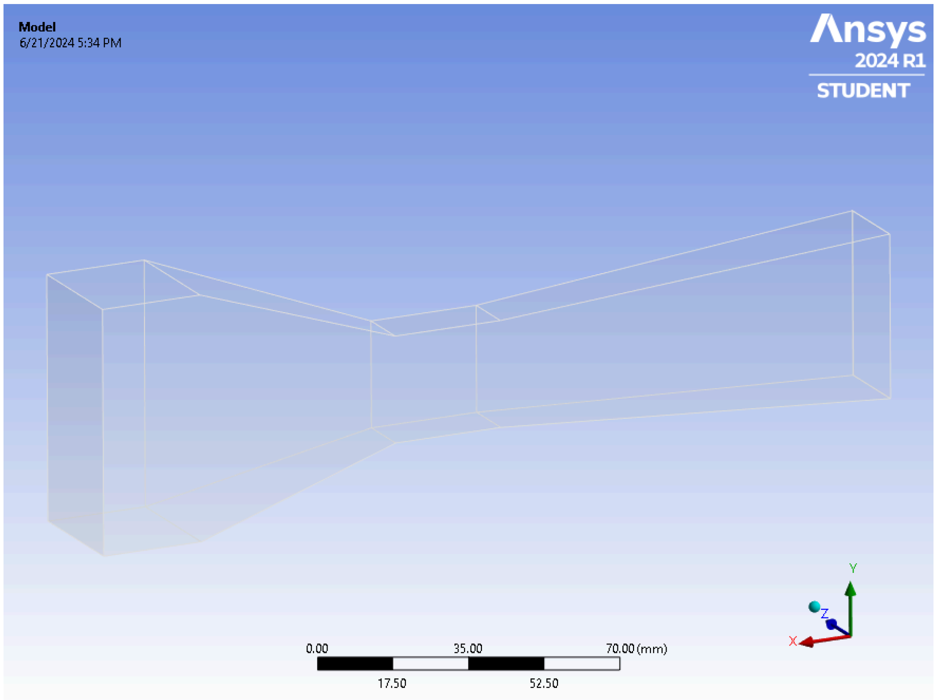

Figure 3-2: Diffuser 40x40mm Geometry

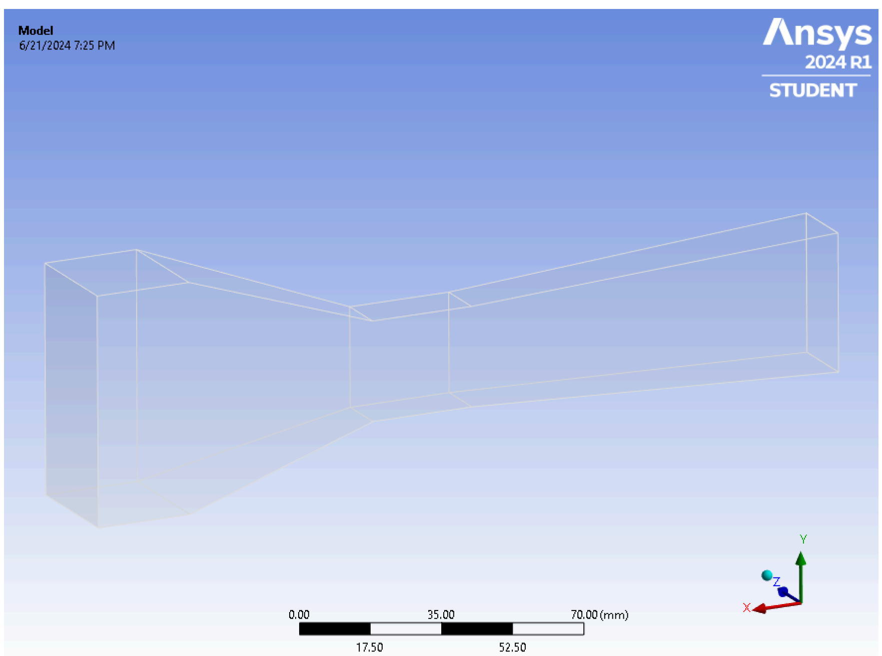

Figure 3-3: Diffuser 36x36mm Geometry

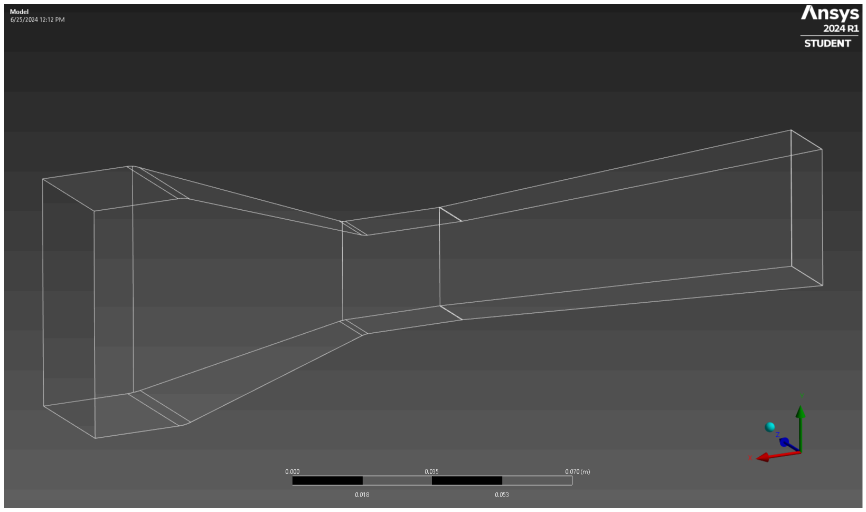

Figure 2-1: Diffuser 36x36mm Geometry with Filets

<u>Tangential Inlet</u>

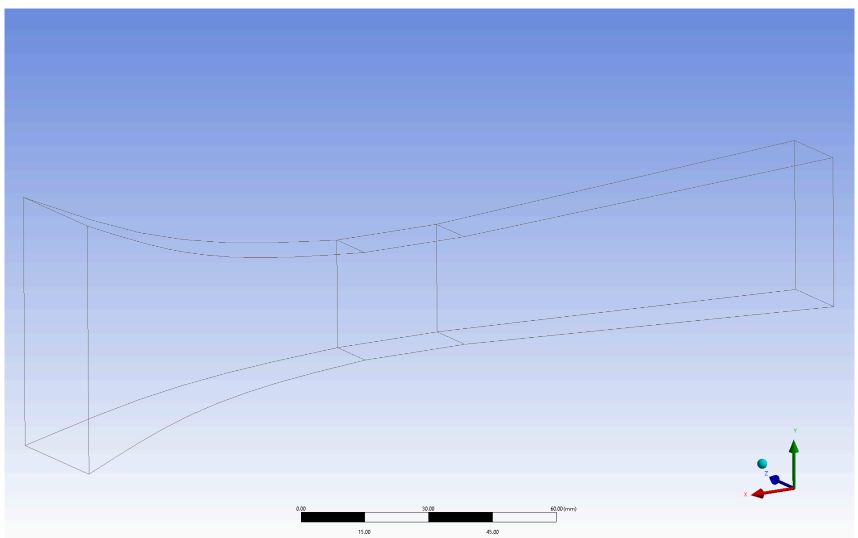

Figure 4-1: Tangential Diffuser 46x46mm Geometry

The tangential inlet design was created to regulate flow-speed from the test section, improve static pressure recovery and reduce pressure loading on the drive system. The flow field within the inlet is influenced by the characteristics of the smooth curvature of the inlet to the test area. Factors such as orientation, size, and curvature angle are major contributors to the air flow entering the test section. To prevent pressure spikes, the inlets length should be increased and the variable curvature radius should have a more gradual change in angle.

Fluid Properties and Boundary Conditions

Reynolds number is used to determine whether the flow is laminar or turbulent. Values under or equivalent to 2300 are laminar while values equal to or over 4000 are turbulent.

The Reynolds number, based on the inlet diameter (width) and velocity is calculated as follows:

- Reynolds Number: $Re = \frac{\rho \bullet V \bullet D}{\mu}$
  - Where ρ is the density, V is the velocity, D is the hydraulic diameter (width), and μ is the viscosity.
  - 46x46 Original Model : $Re = \frac{(1.225 kg/m^3)(.046m)(20m/s)}{(1.7894e{-}05\ kg/ms)} \approx 62982$
  - 36x36 Improved Model : $Re = \frac{(1.225 kg/m^3)(.036m)(20m/s)}{(1.7894e{-}05\ kg/ms)} \approx 49290$

The flows are expected to be turbulent due to the value being significantly over 4000. This is the criteria for the selection in Ansys Fluent of a turbulent flow.

2. Method

Four geometry models were initially designed for testing. After testing for the best diffuser and a new inlet design were completed each modified model and the original model were compared to test for uniform flow in the test section and drag reduction along the walls. The mesh details shown below stays consistent in other testing for the two models.

Diffuser and Inlet

For the diffuser testing three iterations of the wind tunnel were created within the ANSYS Fluent Geometry tool. These three models were then tested to determine if decreasing the angle of attack of the diffuser will affect recirculation, in an attempt to reduce and/remove entirely. Lastly a modified inlet iteration was tested within the ANSYS Fluent Geometry tool. The inlet model was then tested to determine if a smooth tangential curvature of the inlet would reduce recirculation and reduce skin pressure drag. To produce these visuals to identify recirculation zones first a mesh was created with a mesh sizing of 1.6mm (0.0016m). This mesh sizing was determined to be sufficient after mesh independency was confirmed after testing against a mesh sizing of 1.16mm (0.00116m) using the Original 46x46mm geometry, see Figure 1-1. Mesh independence is further shown using an XY Plot below in the uniformity testing section. Note that the number of cells decreasing in the iterations is due to the decrease in overall size of the wind tunnel. Setup followed the selected flow properties and boundary conditions in Figure 0-1, 2, 3, and 4. Residuals were then checked to find convergence, with each version of the wind tunnel going through over 1000 iterations. Velocity Magnitude Vectors plots and Velocity Magnitude Pathlines plots were then created and displayed on the symmetry plane as well as the wall-fluid.

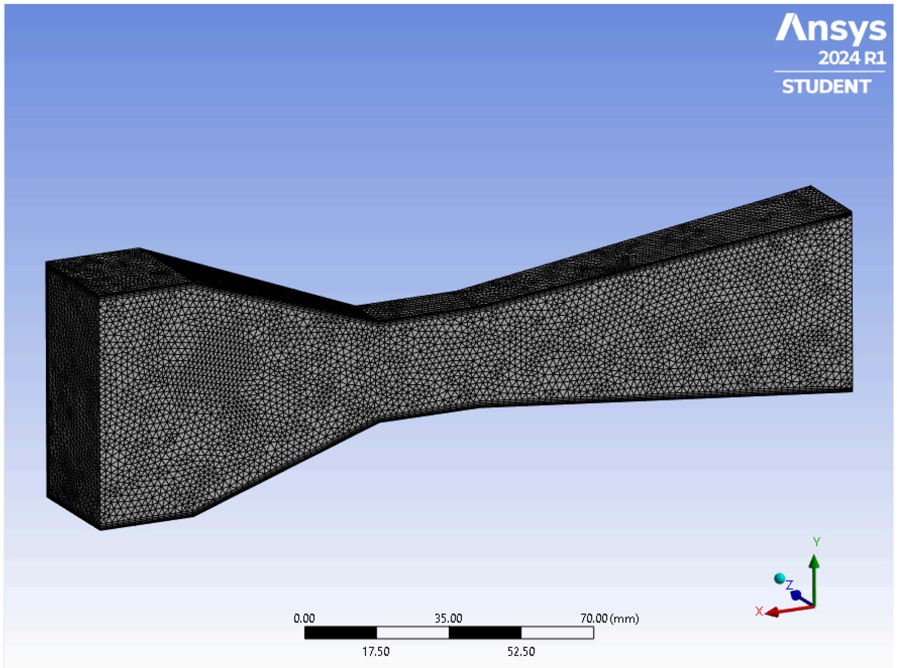

Figure 3-4: Diffuser 46x46mm 1.6mm Mesh - 374,459 Cells

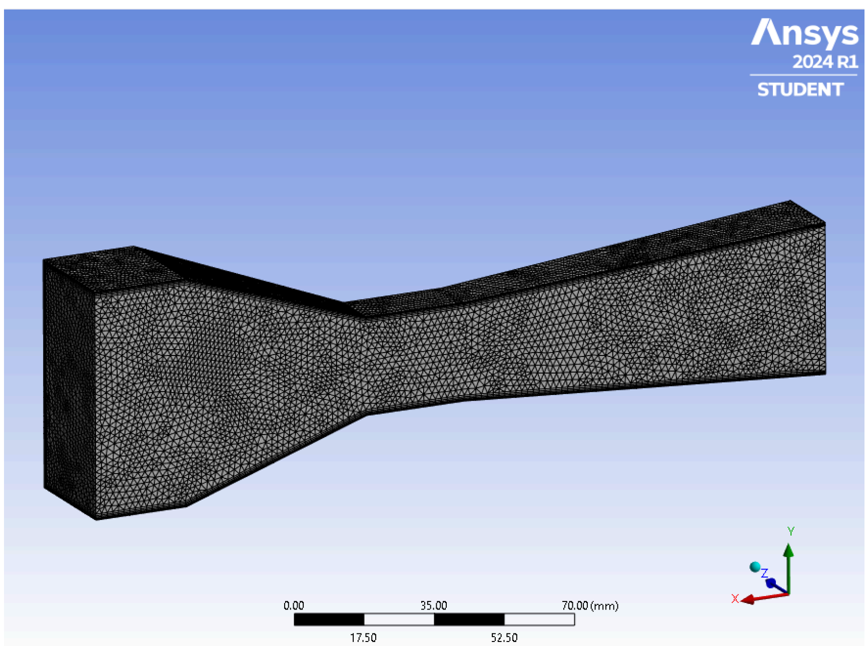

Figure 3-5: Diffuser 40x40mm 1.6mm Mesh - 350,875 Cells

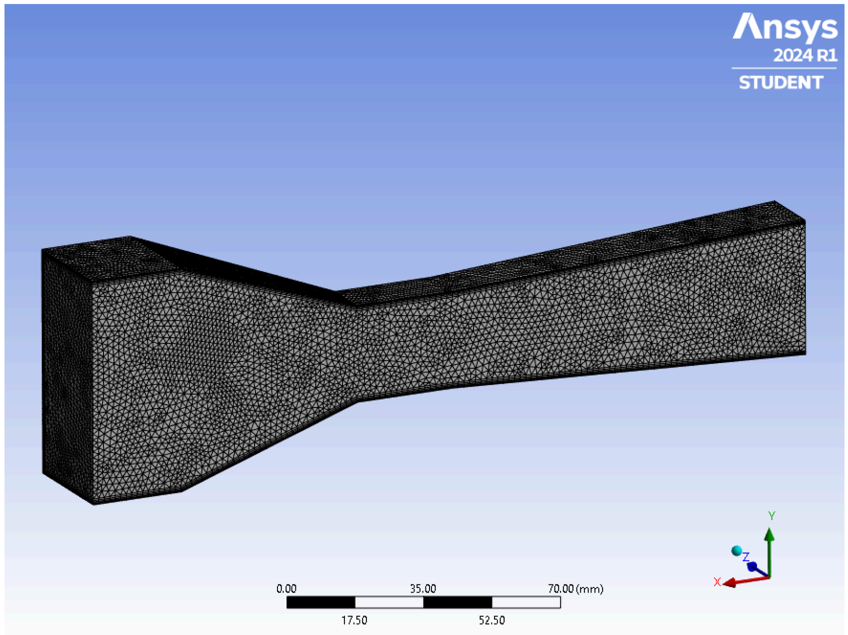

Figure 3-6: Diffuser 36x36mm 1.6mm Mesh - 335,152 Cells

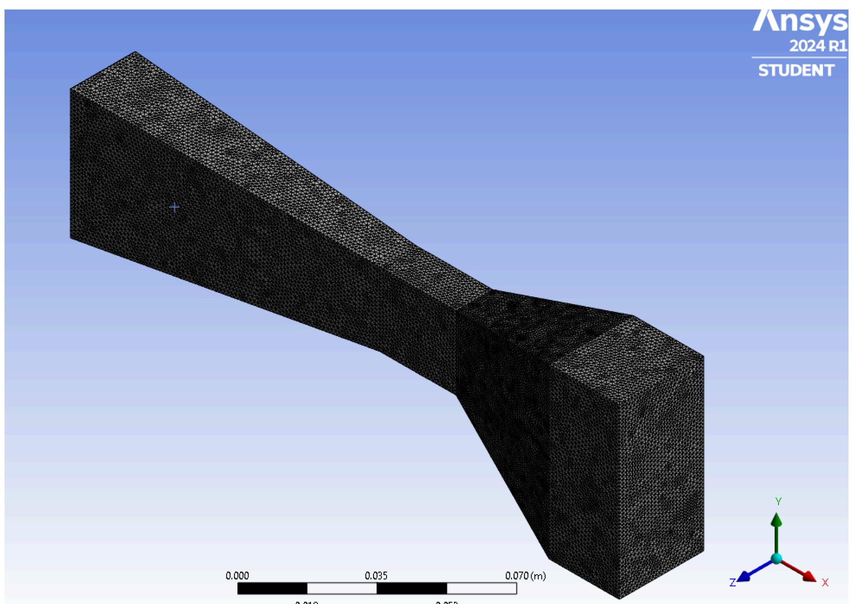

Figure 1-1: Original Model 46x46mm **1.16mm** Mesh - 965,765 (close to 1 million)

Statistics	
Nodes	245442
Elements	985765

Figure 1-2: Fine Mesh Cell Count

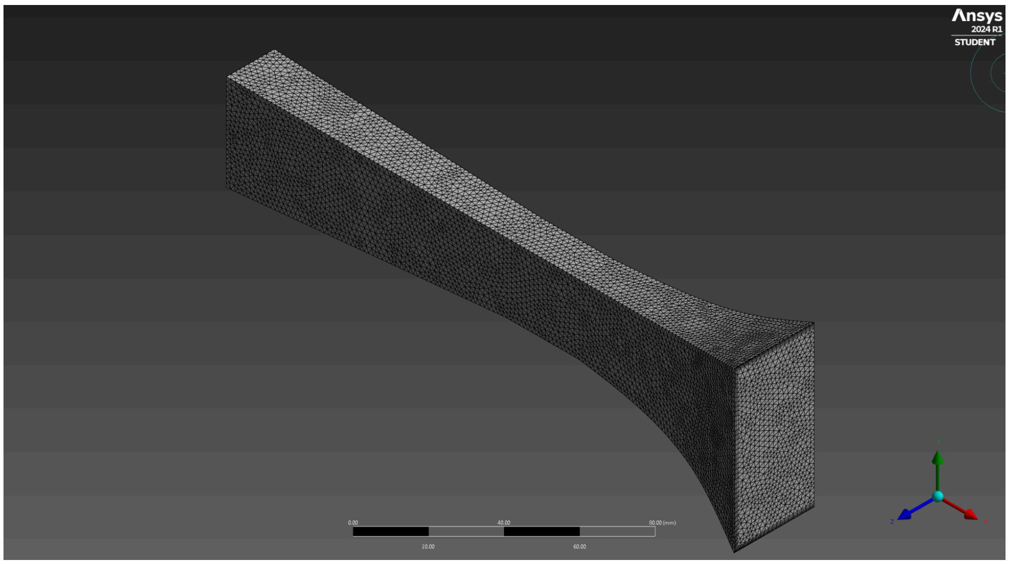

Figure 4-2: Smooth Tangential 46x46mm 1.6mm Mesh - 255,075 Cells

Second order is used for all the spatial discretization methods: Pressure, Momentum, Turbulent Kinetic Energy, and Specific Dissipation Rate due to its higher accuracy. Pressure uses second order while the rest use second order upwind (which includes more data points) allowing for more accuracy and stability.

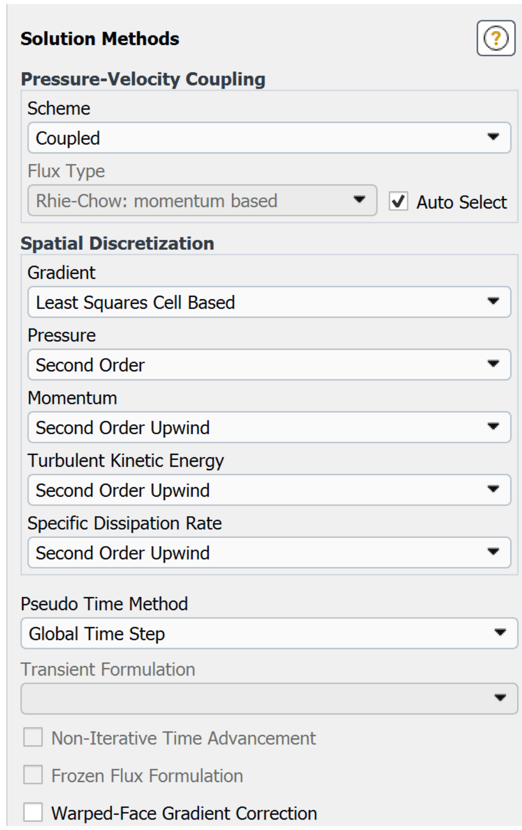

Figure 0-4: Solution Method

<u>Preset and achieved residual values for both geometries</u>

Testing residuals on the wind tunnel models in ANSYS Fluent is crucial for ensuring the accuracy and reliability of computational fluid dynamics (CFD) simulations. Residuals represent the numerical error in the solution and provide insights into the convergence and stability of the simulation. By monitoring residuals, engineers can assess where the simulation has reached a steady state or if further iterations are necessary to achieve accurate results. This iterative process helps in validating the model against experimental data, ensuring that the simulated flow characteristics closely match the real-world conditions. In the tests completed here, comparison with experimental data was not available.

Base Model:

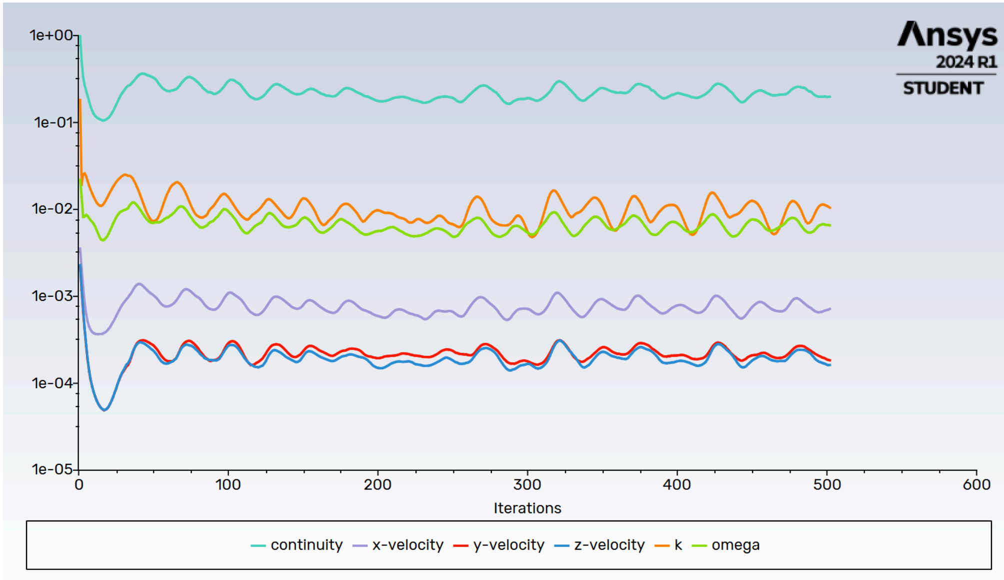

Figure 1-3: 46x46 Residual (Iteration 500)

Figure 1-3 represents the number of iterations ran was initially set to 10,000 allowing as many iterations as needed. At about 500 iterations convergence was believed to be reached. Convergence will be verified using the XY plots below and additional iterations. The residual values below are found at iteration 500.

The achieved residual values:

- continuity: 2e-01
- X-velocity: 7e-04
- Y-velocity: 2e-04
- Z-velocity: 2e-04
- Omega: 7e-03
- K: 1e-02

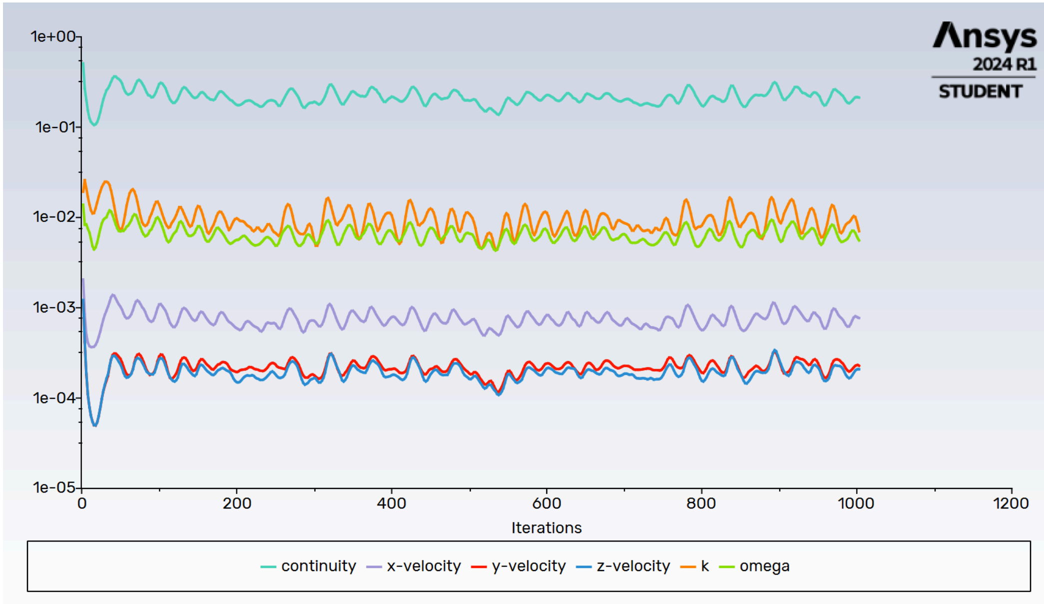

Figure 1-4: 46x46 Residual (Iteration 1000)

Figure 1-4 represents the number of iterations ran was set to an additional 500. The residual values below are found at iteration 1000 to compare with the values at 500 above. There is oscillation, therefore most differences between residual values could be due to the different peaks.

The achieved residual values:

- continuity: 2e-01
- X-velocity: 8e-04
- Y-velocity: 2e-04
- Z-velocity: 2e-04
- Omega: 7e-03
- K: 1e-02

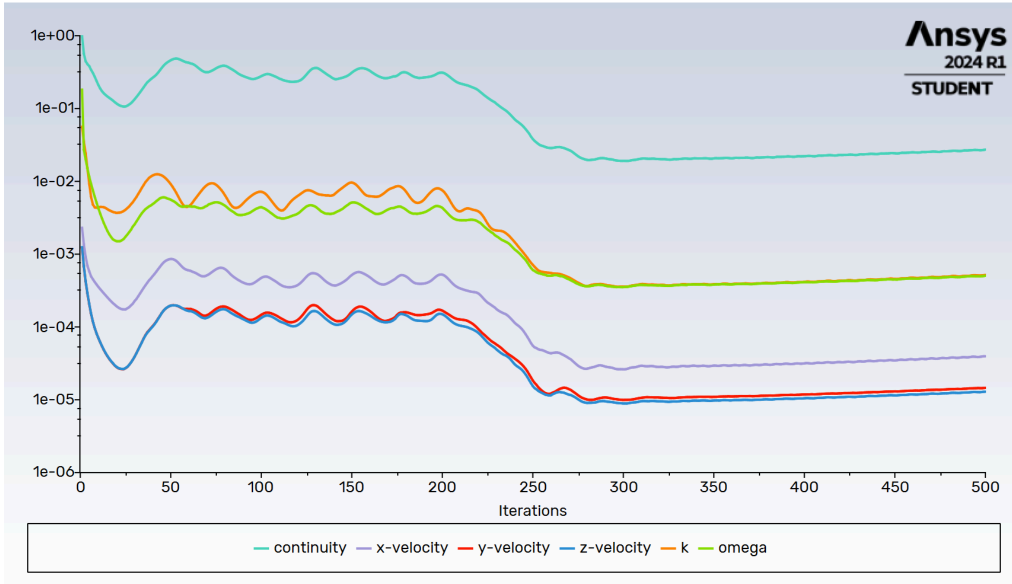

Figure 1-5: 46x46 Residual Fine Mesh(Iteration 1000)

Figure 1-5 represents the number of iterations ran was initially set to 10,000 allowing as many iterations as needed. At about 500 iterations convergence was believed to be reached.

Improved Model:

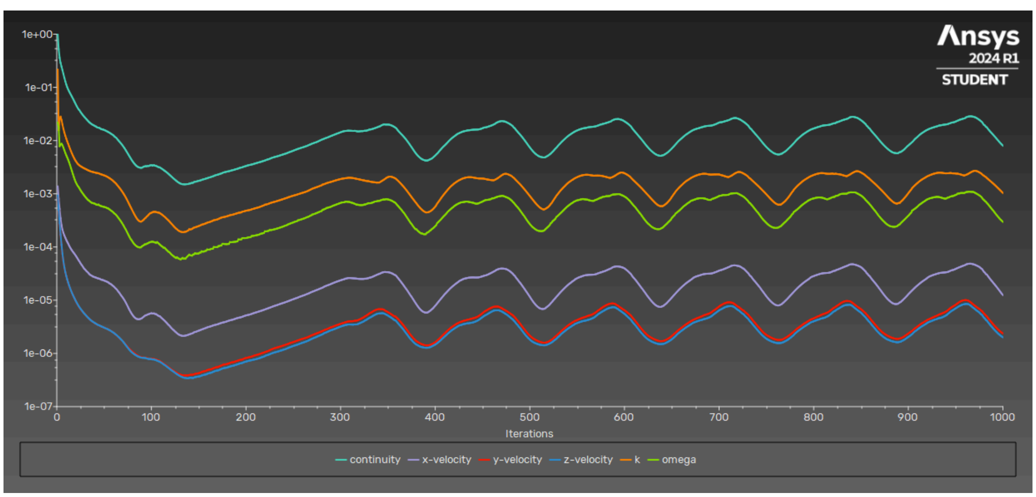

Figure 2-2: 36x36 Residual (Iteration 1000)

Figure 2-2 represents the number of iterations ran to 1000 although at about 400 iterations convergence was reached. Convergence will be verified using the XY plots below. The residual values below are found at iteration 400.

The achieved residual values:

- continuity: 5e-03
- X-velocity: 8e-06
- Y-velocity: 2e-06
- Z-velocity: 1e-06
- Omega: 6e-04
- K: 2e-04

Below are the achieved residual values at iteration 1000.

The achieved residual values:

- continuity: 8e-03
- X-velocity: 1e-05
- Y-velocity: 2e-06
- Z-velocity: 2e-06
- Omega: 1e-03
- K: 3e-04

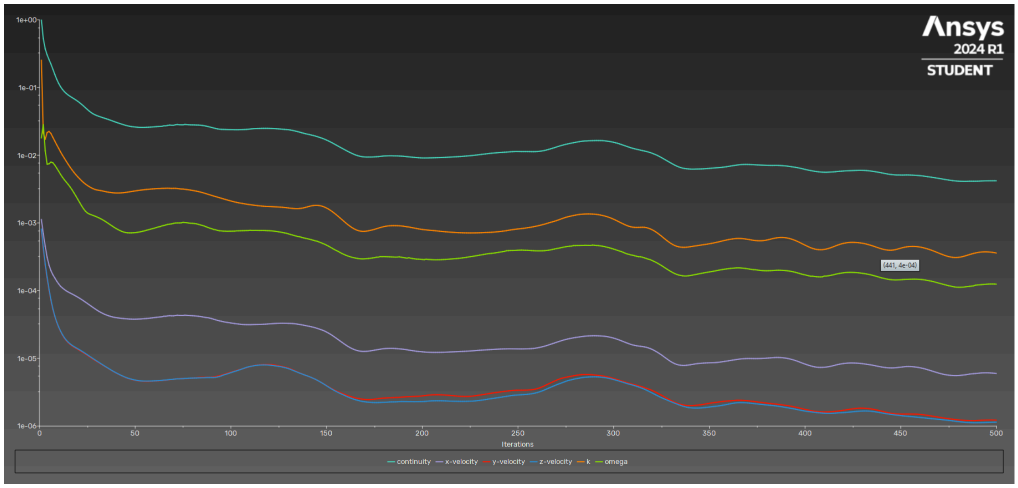

Figure 4-3: Tangential 46x46 Residual (Iteration 500)

Figure 4-3 represents the number of iterations for the Smooth Tangential 46x46 geometry. The number of iterations was initially set to 500 as an initial benchmark to study convergence. At 500 iterations convergence was inferred to not be reached due to the downward trending slope reflected in the residual value plot. To confirm convergence, an additional 500 iterations totaling to 1000 will be run and compared to the initial 500 iteration XY plot below. The residual values below are found at iteration 500.

The achieved residual values:

- continuity: 4e-03
- X-velocity: 5e-06
- Y-velocity: 1e-06
- Z-velocity: 1e-06
- Omega: 1e-04
- K: 3e-04

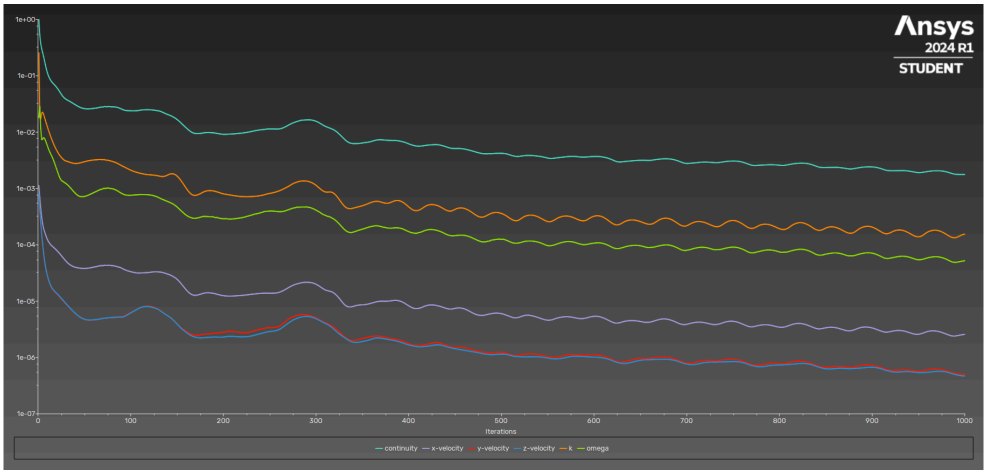

Figure 4-4: Tangential 46x46 Residual (Iteration 1000)

Figure 4-4 represents the second number of iterations for the Smooth Tangential 46x46 geometry. The number of iterations was set to 500 again to achieve a total of 1000 iterations as the second benchmark to study convergence. At 1000 iterations convergence was inferred to once again not be reached due to the downward trending slope reflected in the residual value plot. However, to confirm convergence and create comparisons with other geometric models, comparison of the initial 500 iterations and the 1000 iterations will be used alongside the 500 and 1000 iteration XY plots below. The residual values below are found at 1000 iterations.

The achieved residual values:

- continuity: 1e-03
- X-velocity: 2e-06
- Y-velocity: 5e-07
- Z-velocity: 4e-07
- Omega: 5e-05
- K: 1e-04

3. Results

Recirculation Reduction Testing:

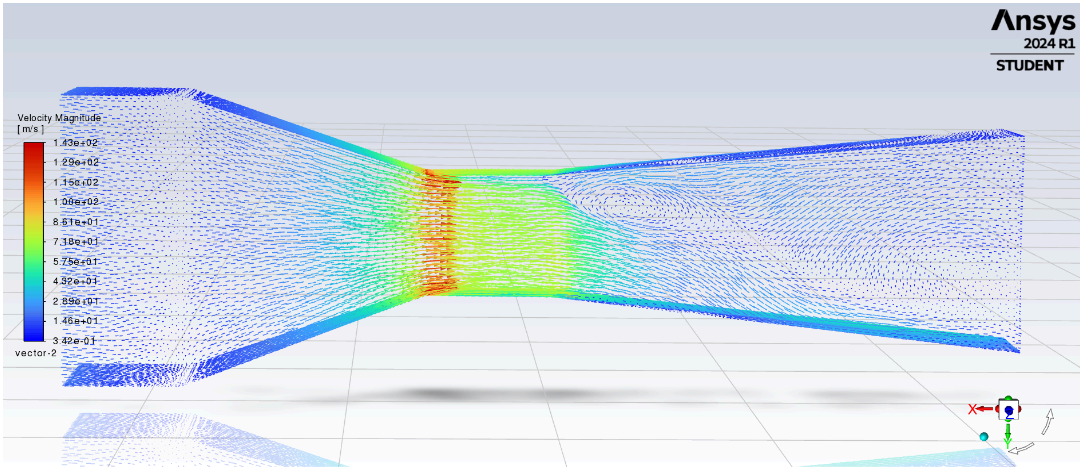

Figure 3-7: Diffuser 46x46mm Velocity Magnitude Vectors Plot

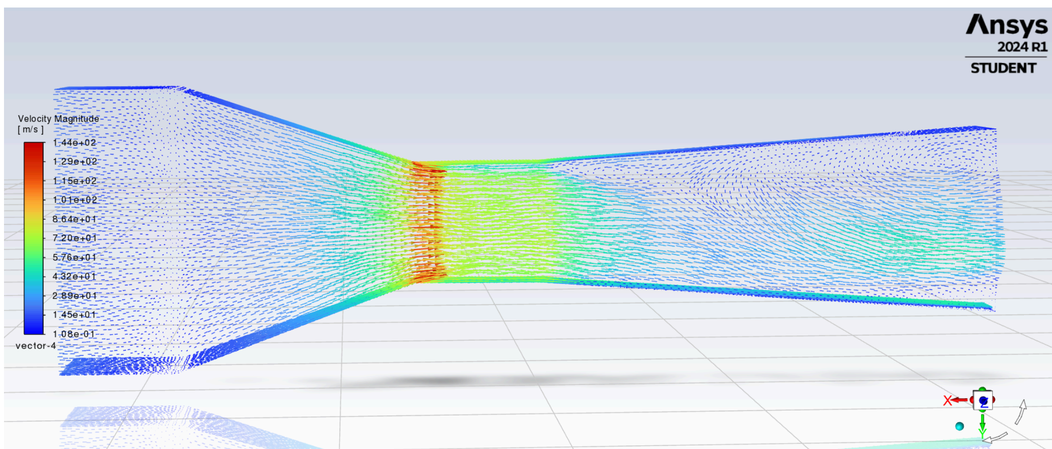

Figure 3-8: Diffuser 40x40mm Velocity Magnitude Vectors Plot

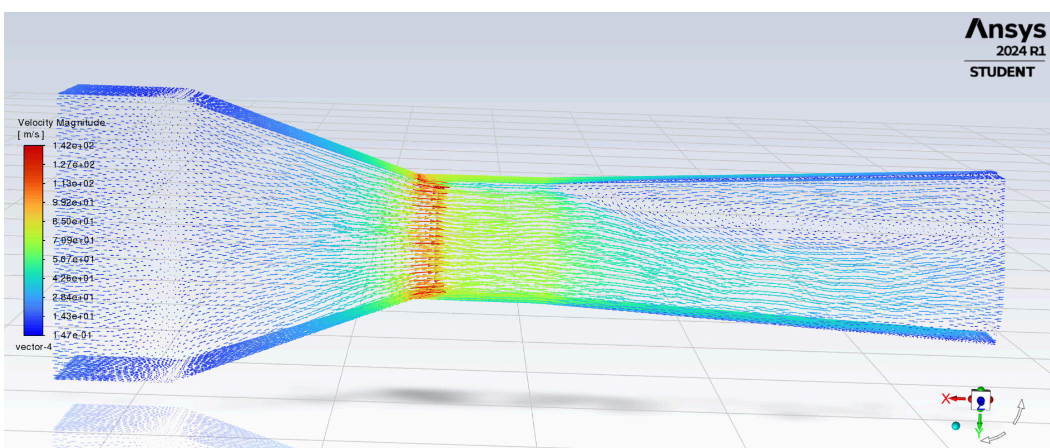

Figure 3-9: Diffuser 36x36mm Velocity Magnitude Vectors Plot

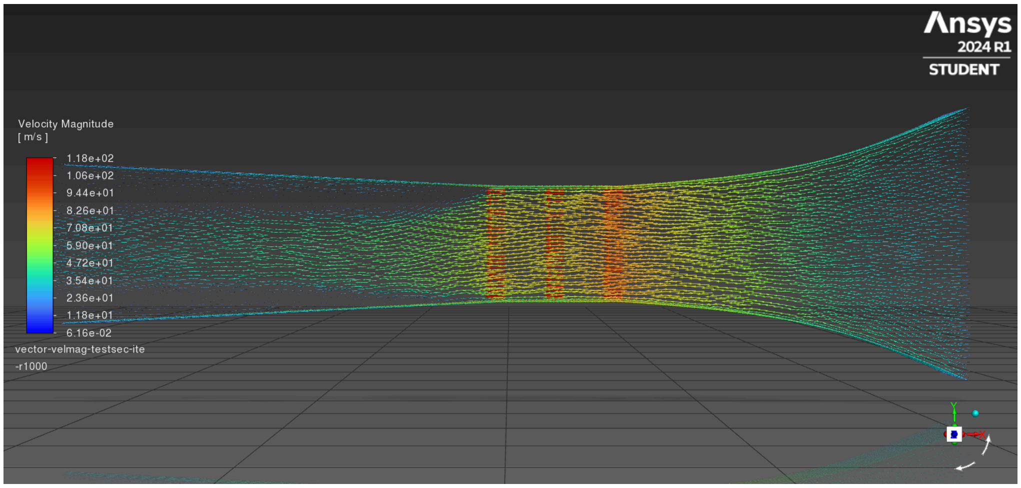

Figure 4-5: Tangential 46x46 Velocity Magnitude Vector Plot

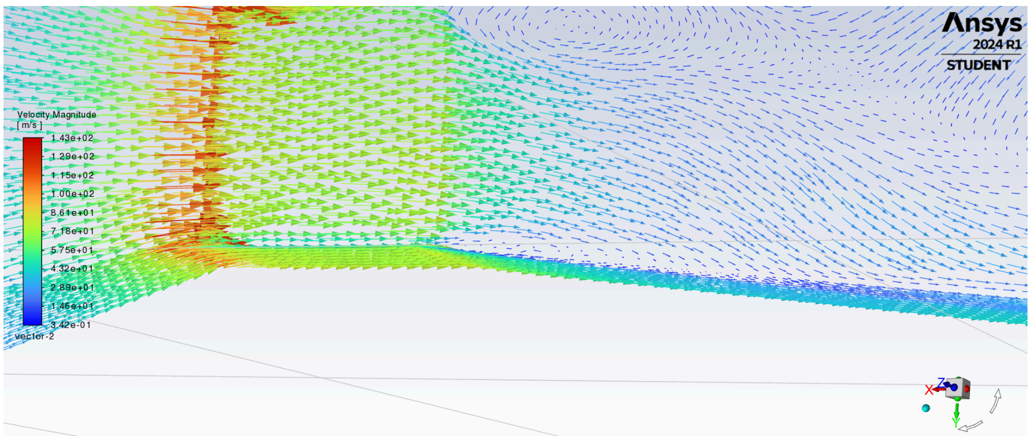

Figure 3-10: Diffuser 46x46mm Velocity Magnitude Vectors Plot

Testing Near Wall Plane

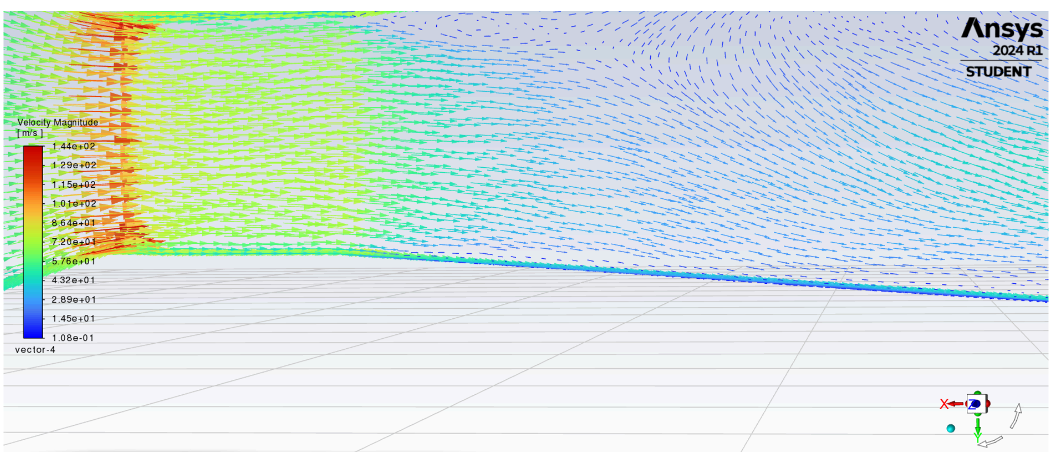

Figure 3-11: Diffuser 40x40mm Velocity Magnitude Vectors Plot

Testing Near Wall Plane

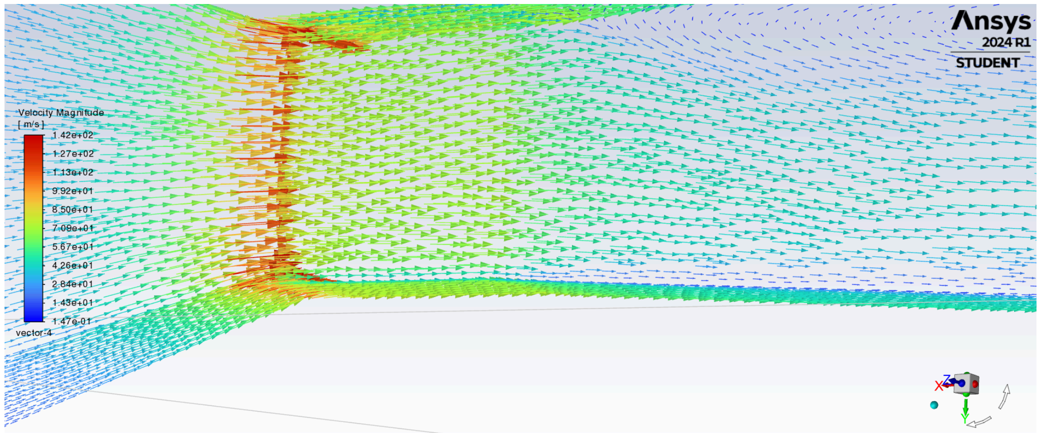

Figure 3-12: Diffuser 36x36mm Velocity Magnitude Vectors Plot

Testing Near Wall Plane

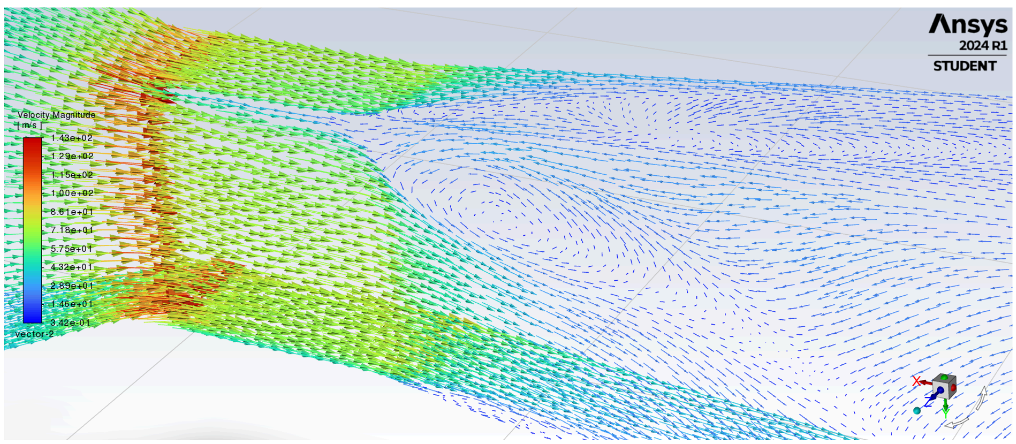

Figure 3-13: Diffuser 46x46mm Velocity Magnitude Vectors Plot

Testing Near Wall Plane

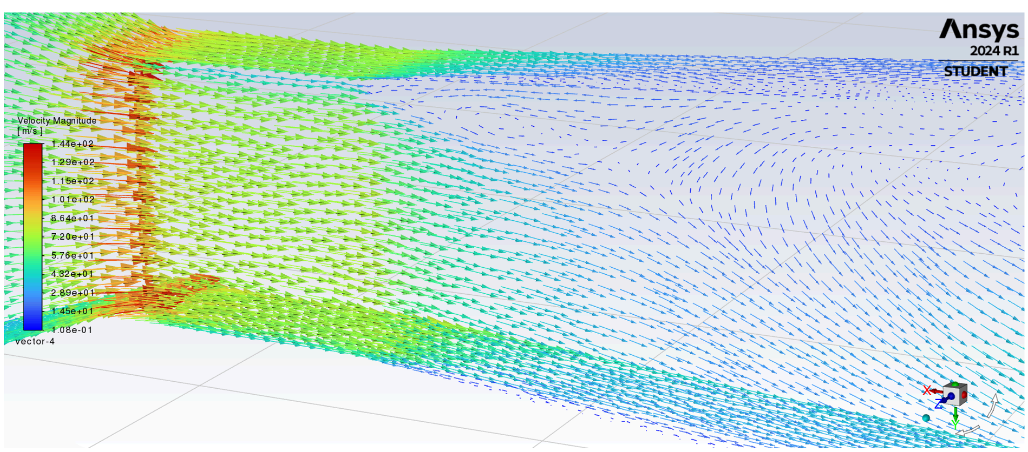

Figure 3-14: Diffuser 40x40mm Velocity Magnitude Vectors Plot

Testing Near Wall Plane

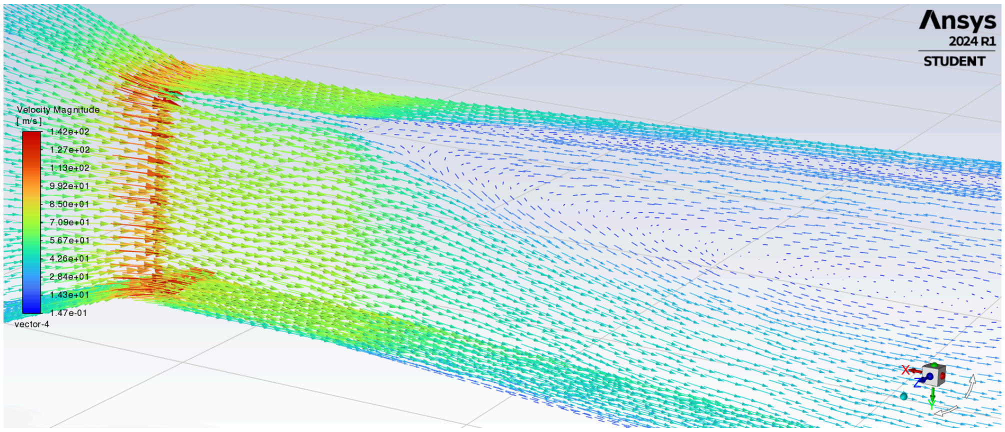

Figure 3-15: Diffuser 36x36mm Velocity Magnitude Vectors Plot

Testing Near Wall Plane

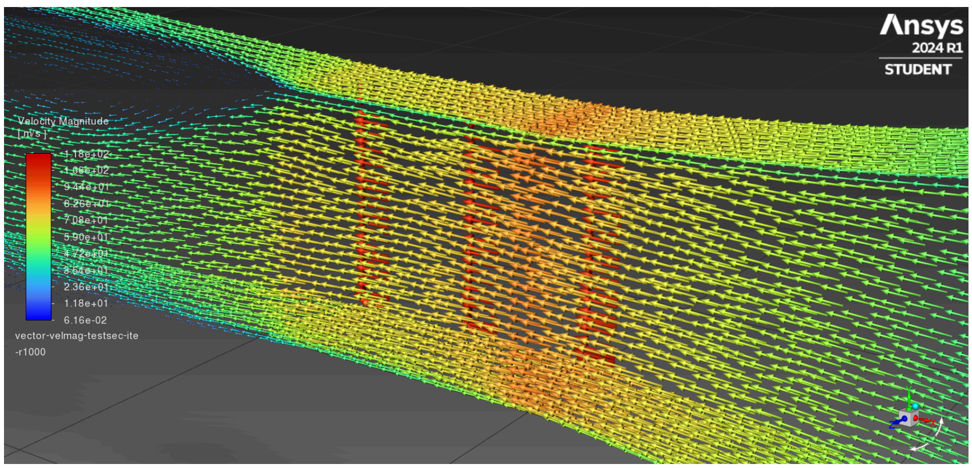

Figure 4-6: Tangential 46x46 Velocity Magnitude Vector Plot

Testing Near Wall Plane

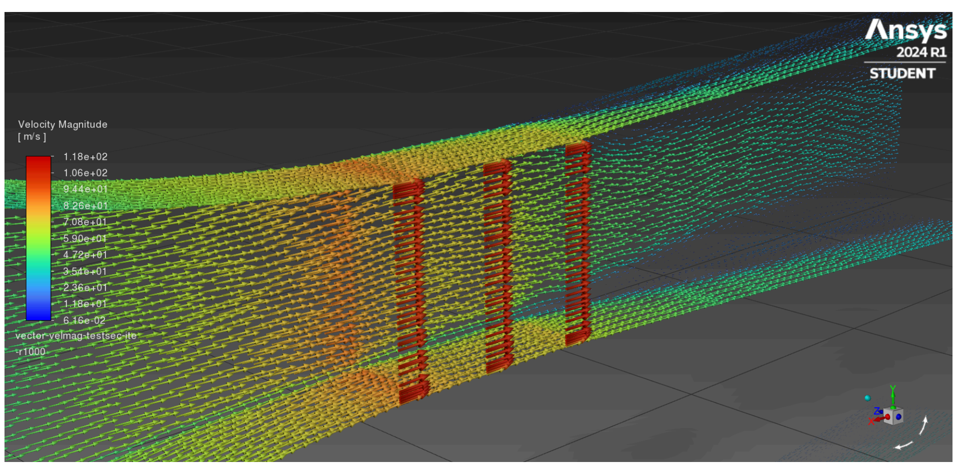

Figure 4-7: Tangential 46x46 Velocity Magnitude Vector Plot

Testing Near Wall Plane

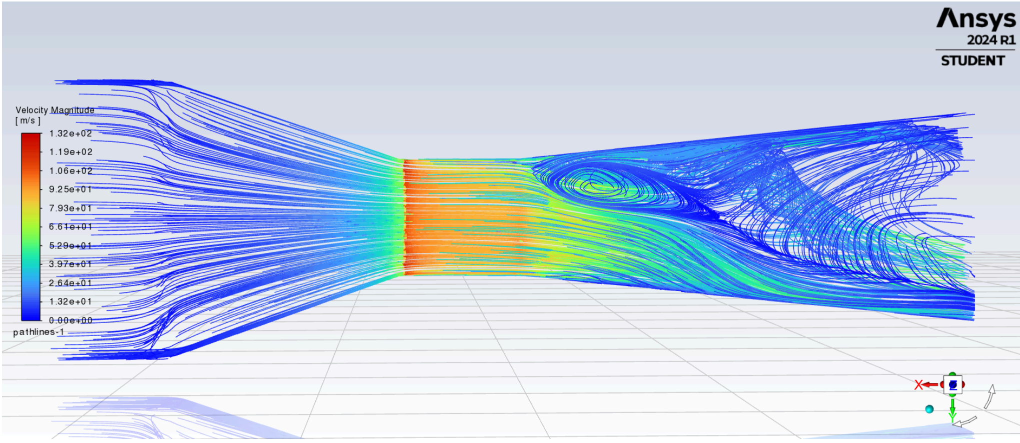

Figure 3-16: Diffuser 46x46mm Velocity Magnitude Pathlines Plot

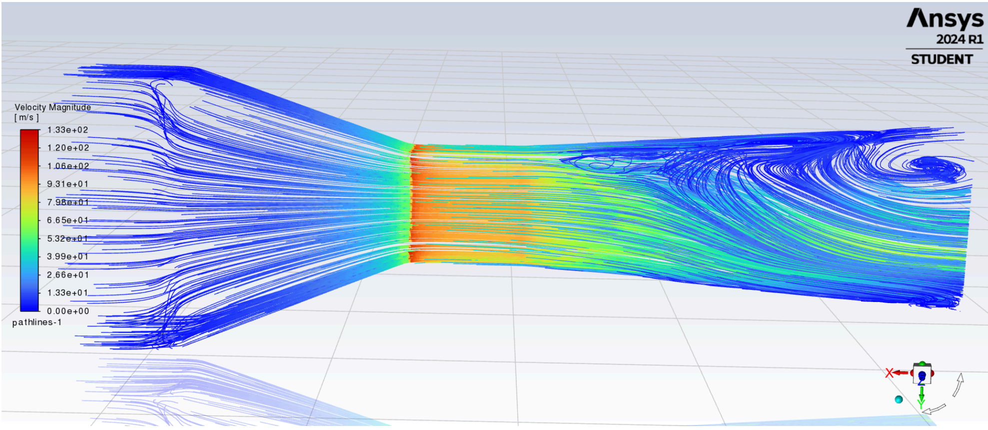

Figure 3-17: Diffuser 40x40mm Velocity Magnitude Pathlines Plot

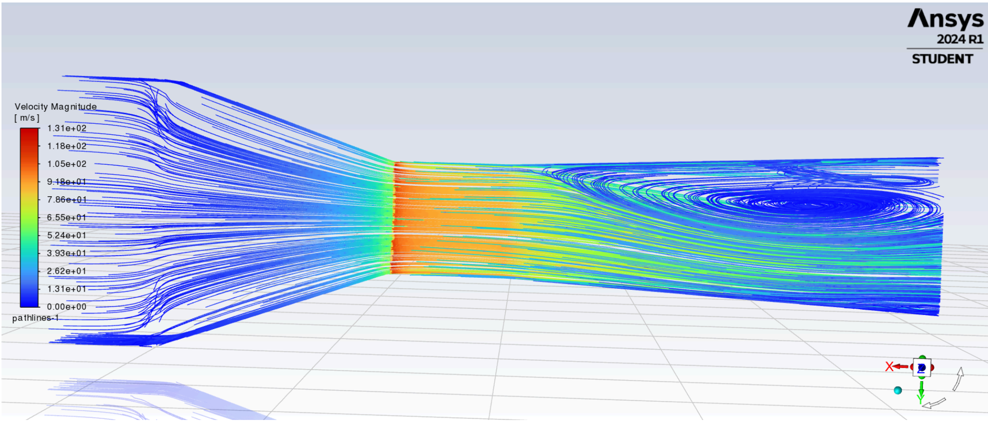

Figure 3-18: Diffuser 36x36mm Velocity Magnitude Pathlines Plot

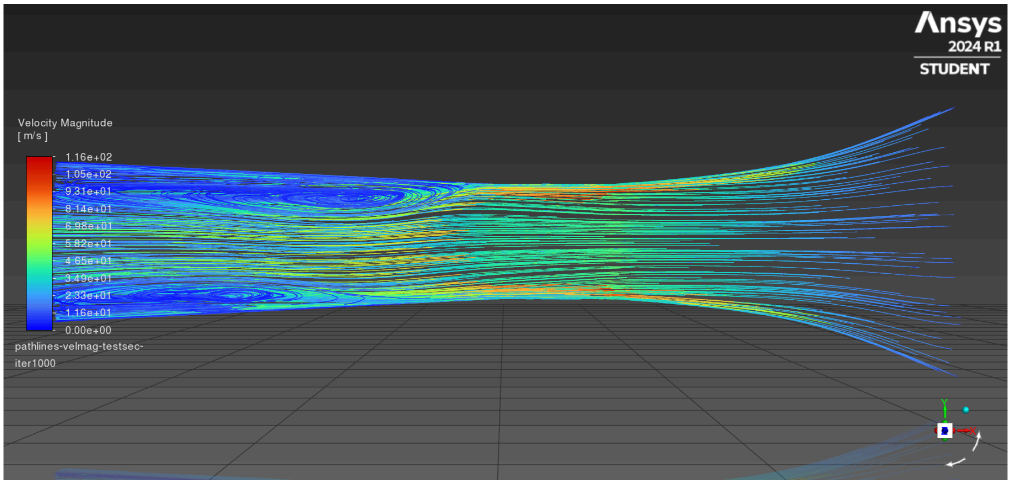

Figure 4-7: Tangential 46x46 Velocity Magnitude Pathlines Plot

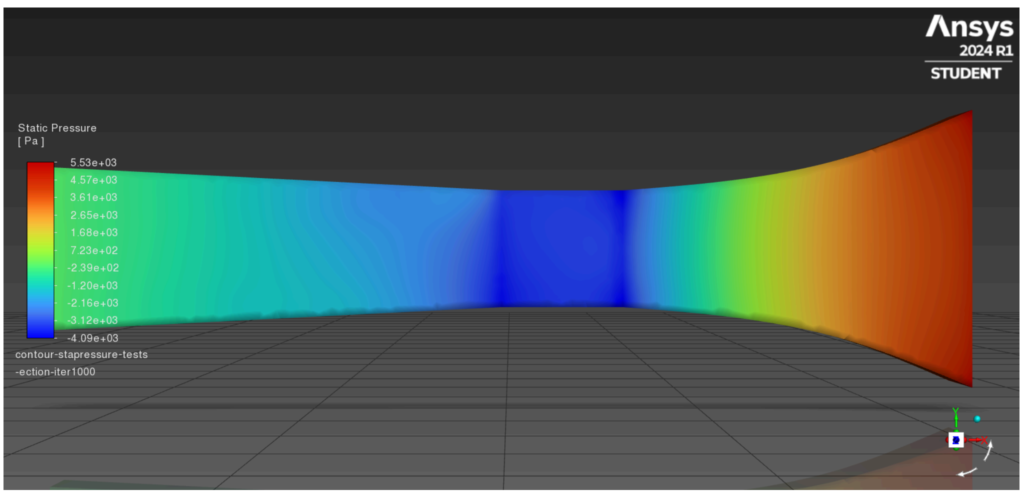

Figure 4-8: Tangential 46x46 Static Pressure Contour Plot

Uniform Flow Testing Through Test Section

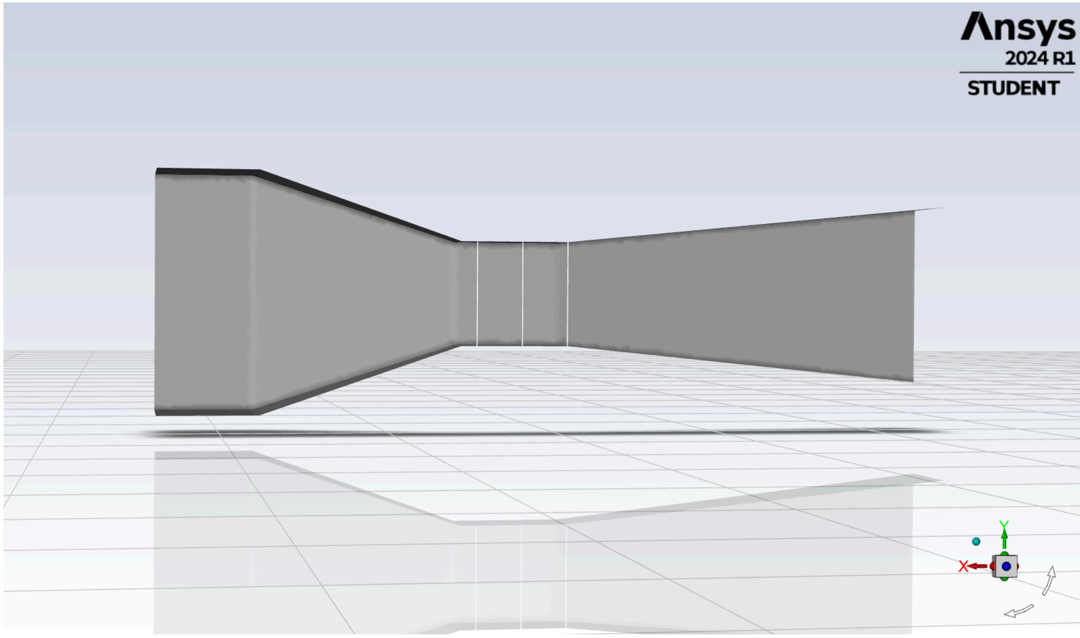

Figure 1-6: Lines Created for XY Plots at Test Section (Original Model, 46x46)

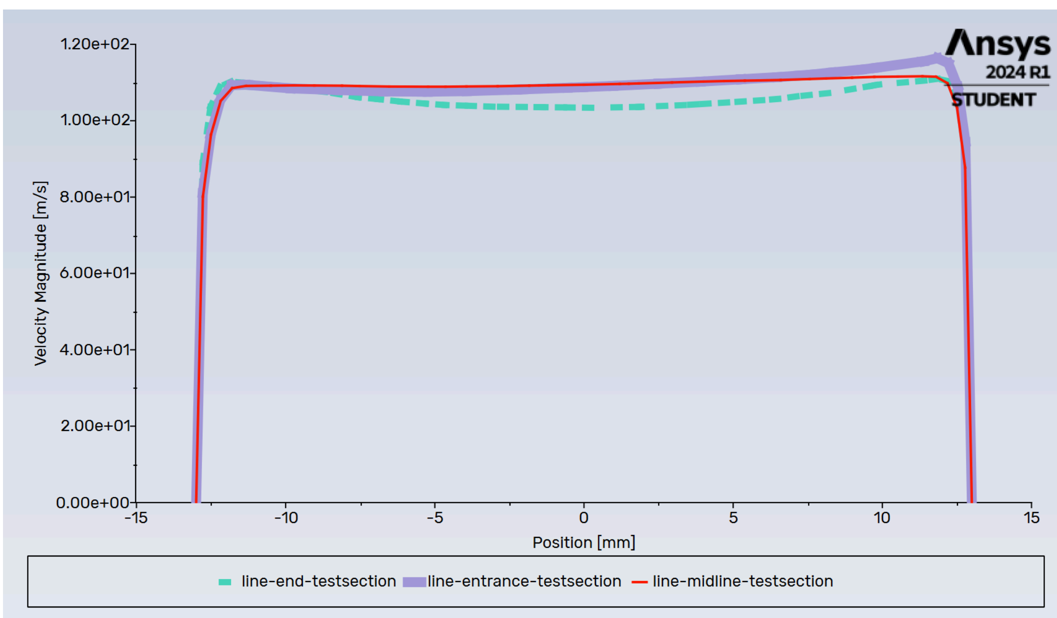

Figure 1-7: Velocity Magnitude XY Plot at 3 Test Section Lines (Iteration 500, Original Model, 46x46)

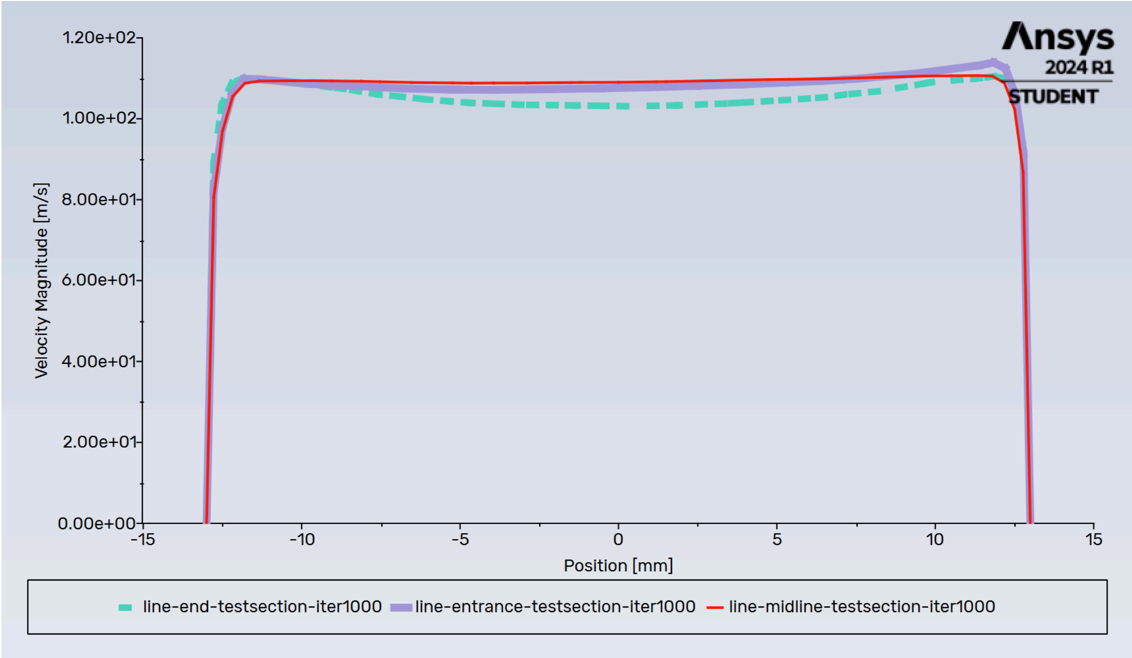

Figure 1-8: Velocity Magnitude XY Plot at 3 Test Section Lines (Iteration 1000, Original Model 46x46)

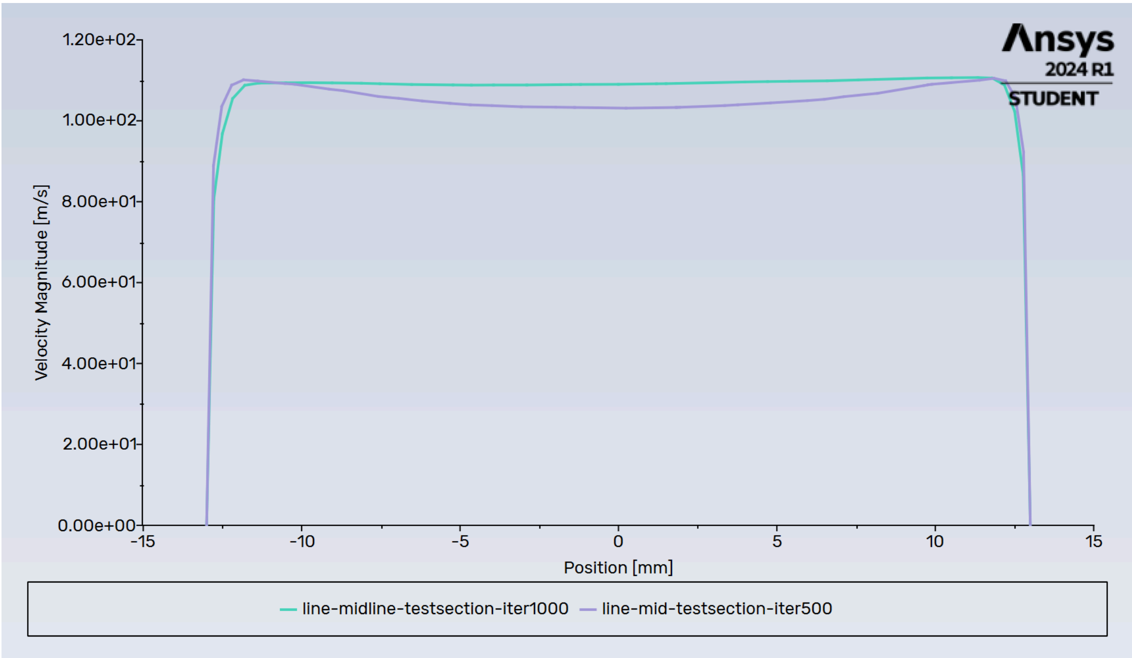

Figure 1-9: Velocity Magnitude Convergence test (Iterations 500 & 1000, Original Model 46x46)

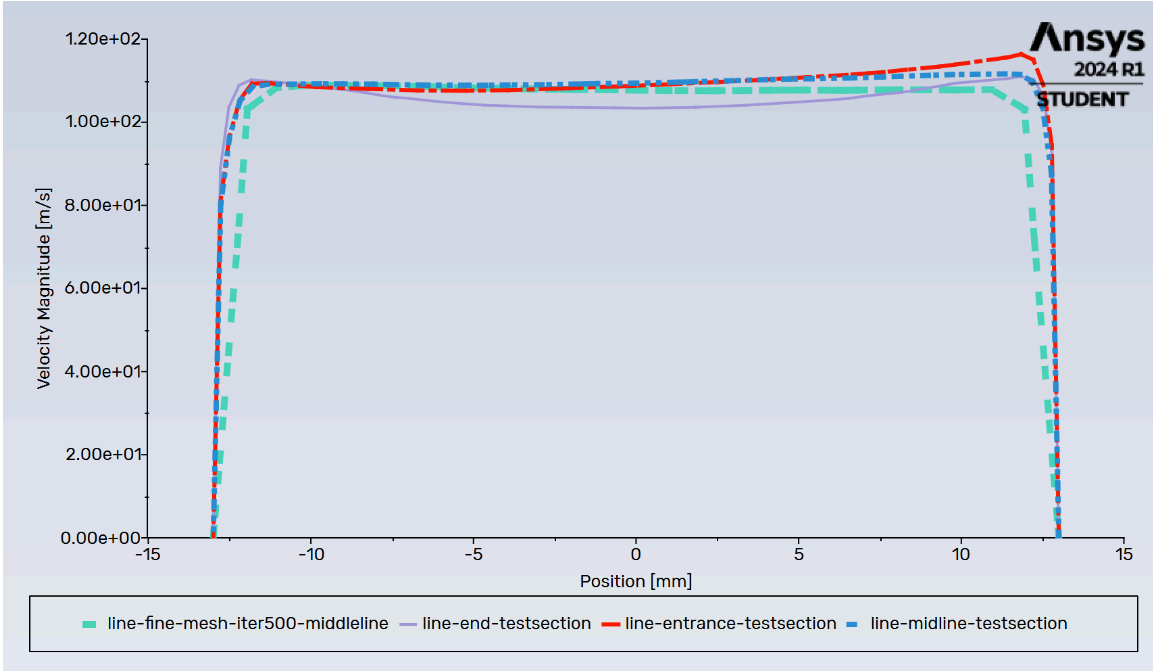

Figure 1-10: Mesh Comparison: Velocity Magnitude XY Plot of Mesh and Fine Mesh (Iterations 500, Original Model 46x46)

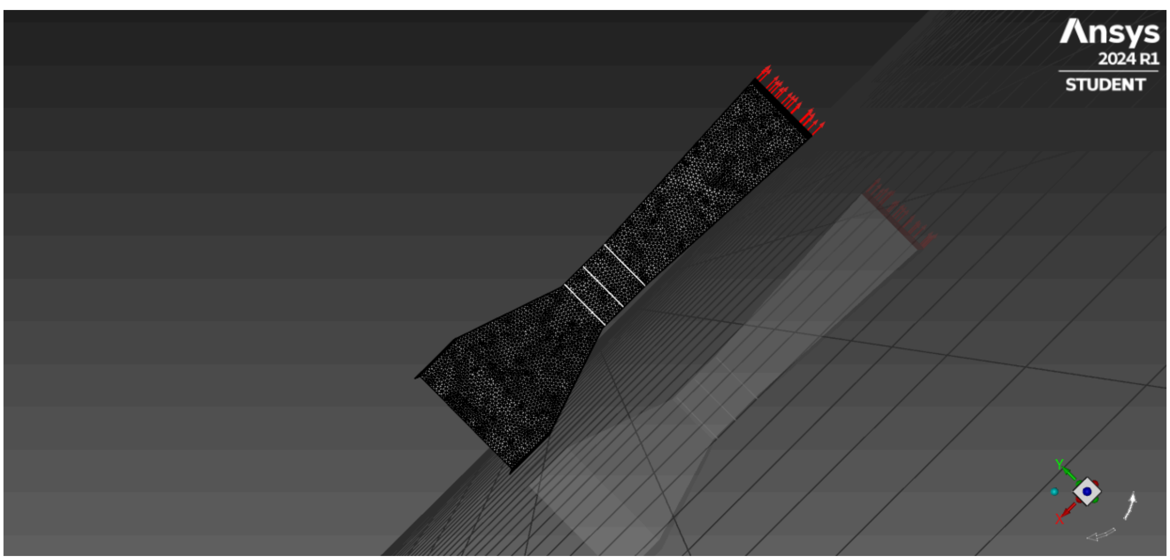

Figure 2-3. Lines Created for XY Plots at Test Section (Modified Model, 36x36)

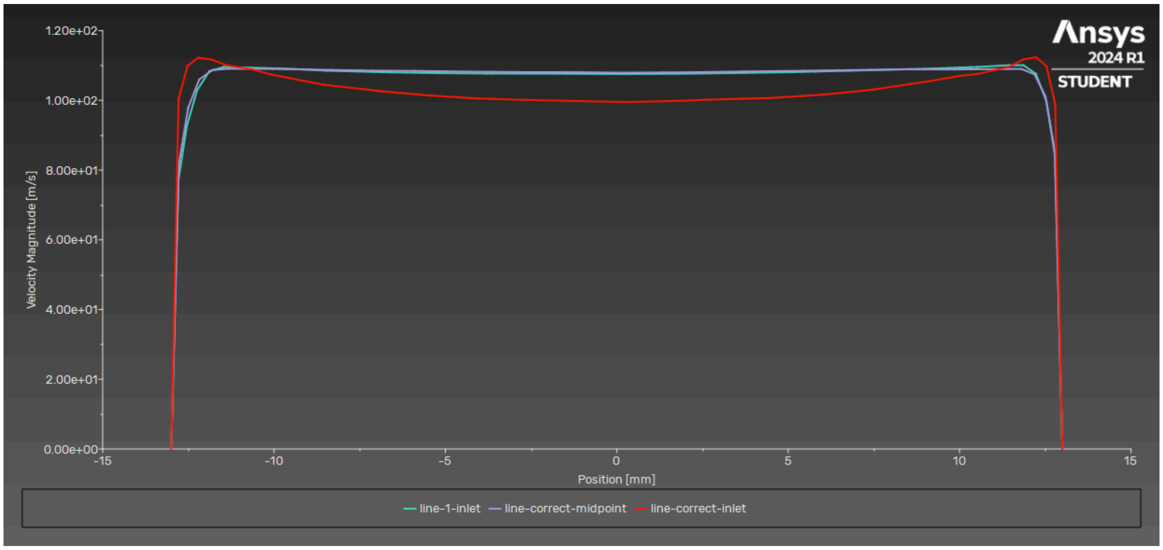

Figure 2-4: Velocity Magnitude XY Plot at 3 Test Section Lines (Modified Model (without filets), 36x36, iteration 1000)

(line 1 inlet is actually outlet)

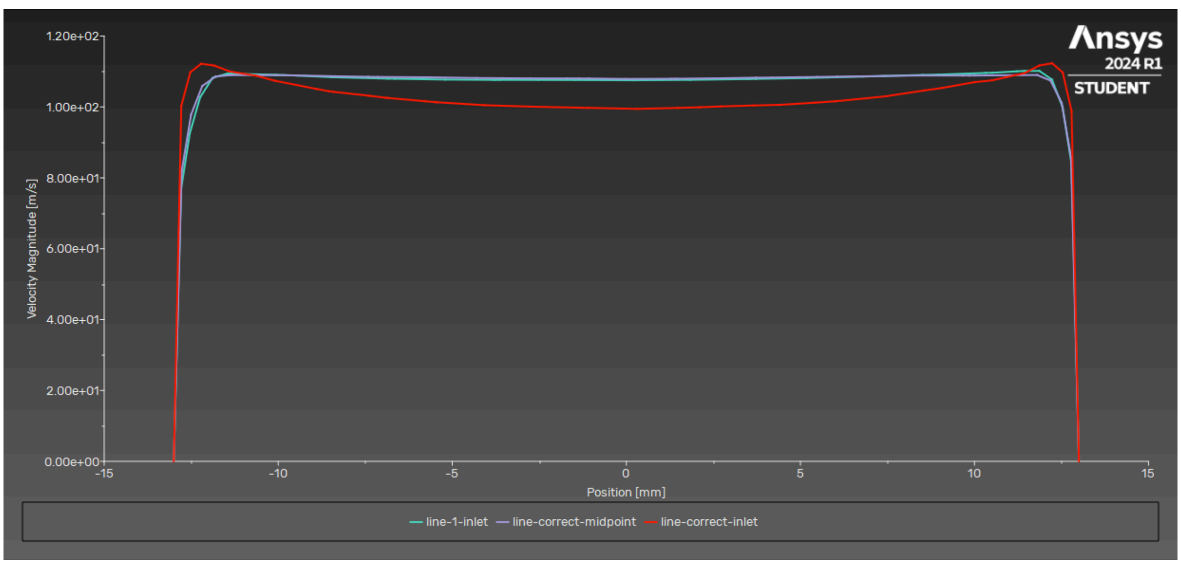

Figure 2-5: Velocity Magnitude XY Plot at 3 Test Section Lines (Modified Model (without filets), 36x36, iteration 400)

(line 1 inlet is actually outlet)

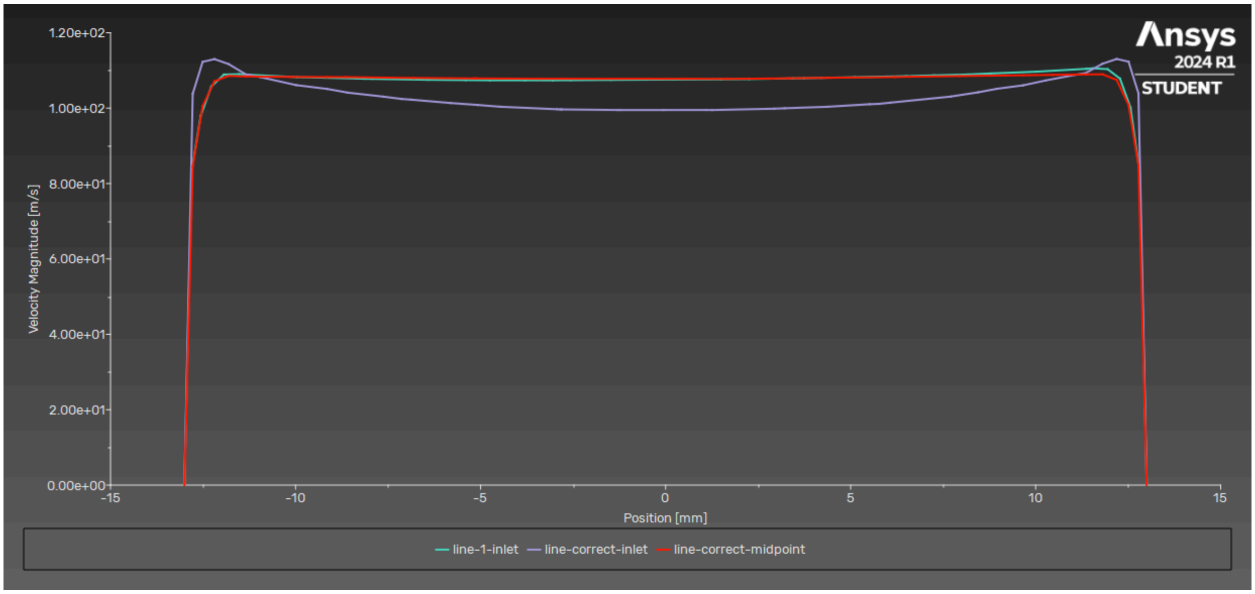

Figure 2-6: Velocity Magnitude XY Plot at 3 Test Section Lines (Modified Model (with filets), 36x36, iteration 400)

(line 1 inlet is actually outlet)

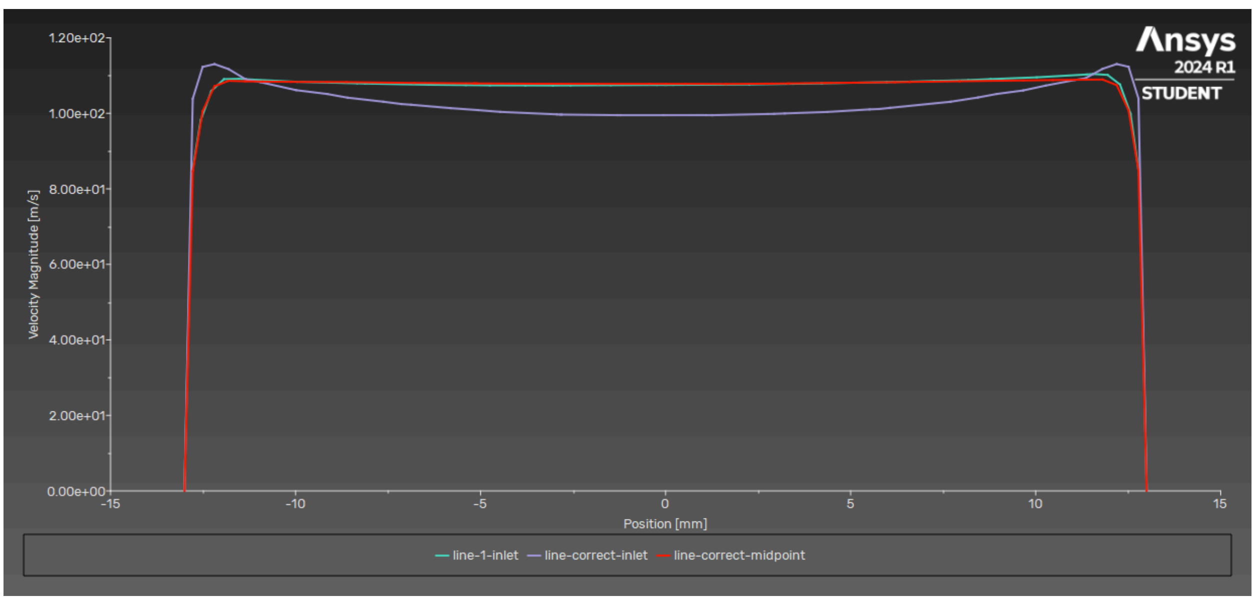

Figure 2-7: Velocity Magnitude XY Plot at 3 Test Section Lines (Modified Model (with filets), 36x36, iteration 1000)

(line 1 inlet is actually outlet)

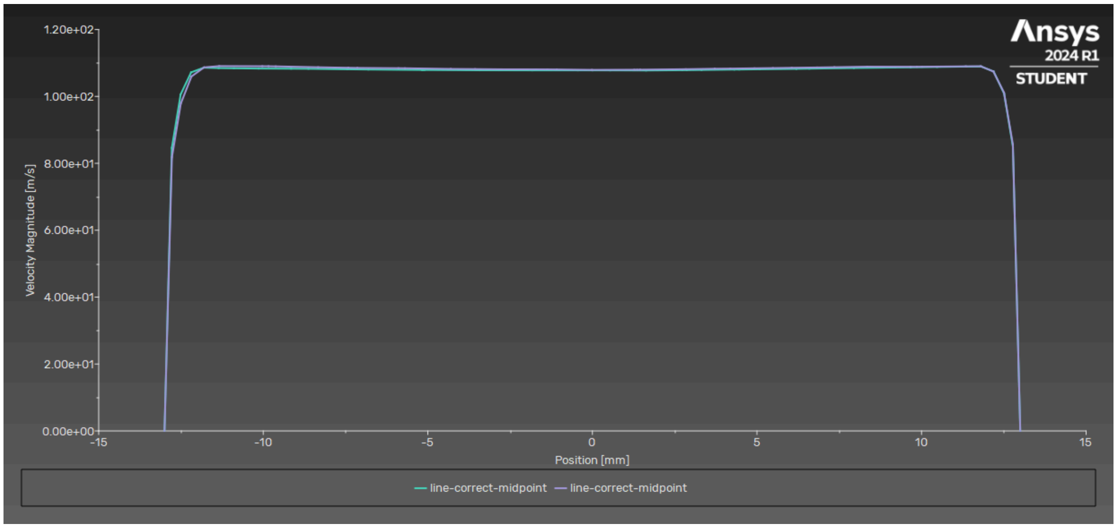

Figure 2-8: Velocity Magnitude XY Plot comparing Uniformity Between Geometry with Filets and Without. (36x36, iteration 1000)

(purple represents with filets)

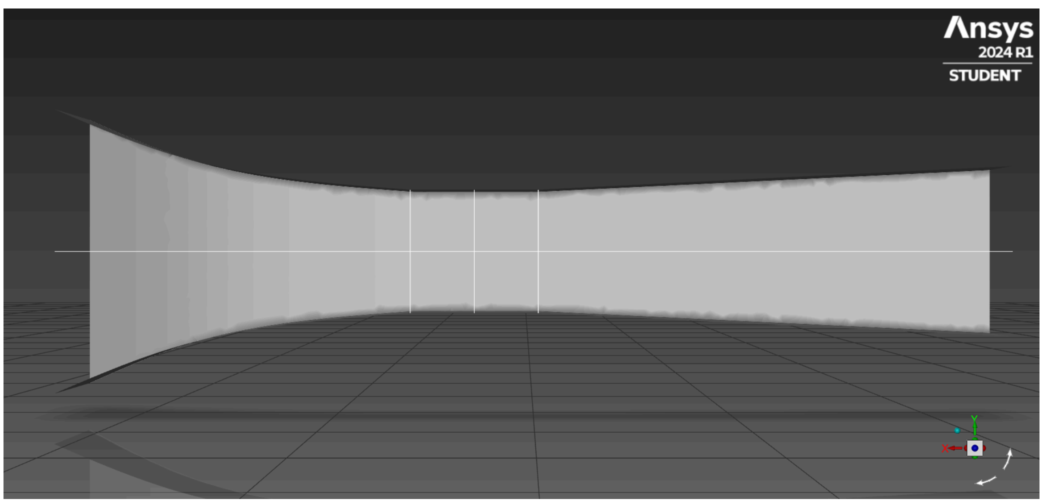

Figure 4-7: Lines Created for XY Plots at Test Section (Tangential 46x46, Middle Test Section Line is present but faintly colored)

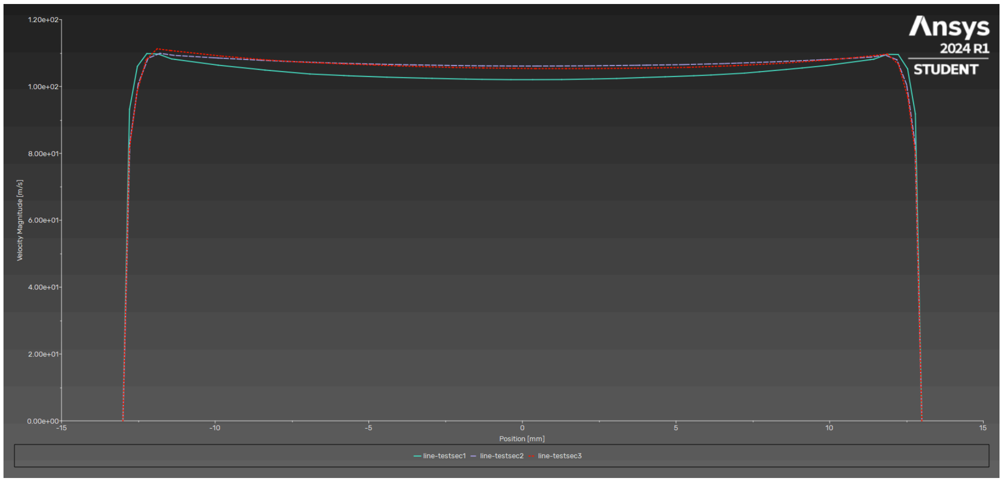

Figure 4-8: Velocity Magnitude XY Plot at 3 Test Section Lines (Tangential 46x46, Iteration 500)

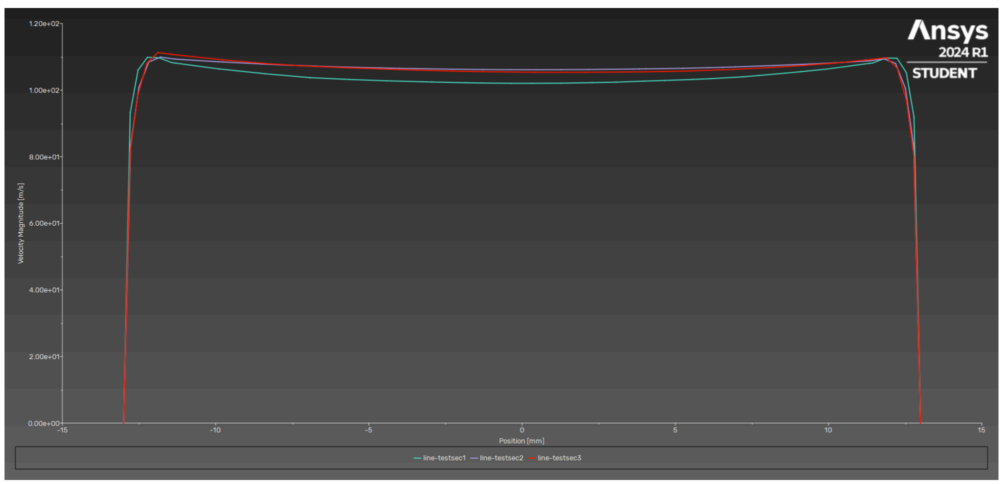

Figure 4-9: Velocity Magnitude XY Plot at 3 Test Section Lines (Tangential 46x46, Iteration 1000)

4. Discussions

Diffuser

Figures 3-7, 8, and 9 are Velocity Vectors plots for the 3 different size diffusers that were tested, showing a complete look at the wind tunnel. Figures 3-10, 11, 12, 13, 14, and 15 are a closer look at near wall planes of the wind tunnel that produced recirculation zones when testing. Figures 3-16, 17, and 18 are Velocity Pathlines plots providing a more dynamic look at the recirculation zone created from each size diffuser. The 46x46mm diffuser displayed the largest recirculation area, even causing airflow to whip back towards the test area within the diffuser. Looking at the Pathlines plots, as the diffuser exit area decreased from 46x46mm to 40x40mm to 36x36mm the recirculation area after the test area shrinks in vertical height, as well as moving further away from the testing area. Future tests could be run to determine if increasing the length of the diffuser or creating a negative attack angle after the test area could reduce recirculation zones. Although a longer diffuser provides greater pressure recovery, size limitations often constrain the diffuser length and height. [3]

Uniformity in Test Section

A uniform flow is ideal for the accuracy and even distribution though the test section of a wind tunnel. The method used to test for a uniform flow through the test section was to create 3 lines perpendicular to the flow (Figure 1-6, 2-3, 4-7) and perform XY plots of the velocity magnitude to show the relationship of the velocity force (N) to the position (mm) across the line or cross section plan. The three plots were made at iteration 500 and 1000 (Figures 1-7, 1-8) and compared to see similarity between them. An XY plot of each iteration is compared to test for convergence (Figure 1-9). A velocity magnitude XY Plot of the fine mesh was compared to the plot of the more coarse mesh (using the middle line of the test section for both) in (Figure 1-10). Along with this, Figure 2-4,5 represents the base 36 x 36 model at 400 and 1000 iterations. These Results are in comparison to the modified 36 x 36 (Figure 2-6,7). Figure 2-8 directly compares the results from the XY plot of the two different models. Lastly, Figure 4-8, and 9 displays a velocity magnitude XY Plot for comparison with other geometric plots.

Tangential Inlet Model

For Figure 4-5, 6, 7, and 8 various plots are taken using a color gradient legend. In Figure 4-5, 6, and 7 the velocity magnitude is shown via a velocity magnitude vector plot and velocity magnitude pathlines plot. Within those plots the tangential inlet effectively smoothens the rate of change for velocity within the whole system compared to the diffuser designs shown above. Within Figure 4-8, a contour plot of static pressure along the walls of the system shows a drastic reduction in pressure as the airflow approaches and leaves the test section. However, in the post-test section the system experiences medium sized recirculation zones at each corner within the diffuser which is a direct increase in the number of recirculation zones that exist in the system. In exchange, the medium sized recirculation zones are smaller in overall size compared to its diffuser modified counterparts. In the future, adding fillets in the corners of the wind tunnel, extending the length of the inlet can further improve the qualities of the inlet design.

Additional Approach

The wind tunnel design for this project is a solid model although another method was explored. In order to set up the space inside the hollow tunnel as the fluid domain the boolean tool was used. There is a solid rectangle surrounding the tunnel, a shell (tunnel walls), and the fluid (area inside).

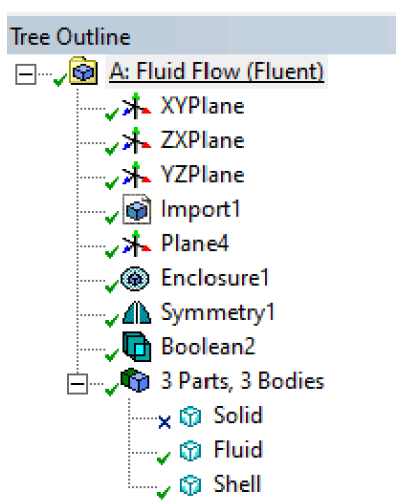

Figure 1-11. Outline Showing Different Parts Labeled

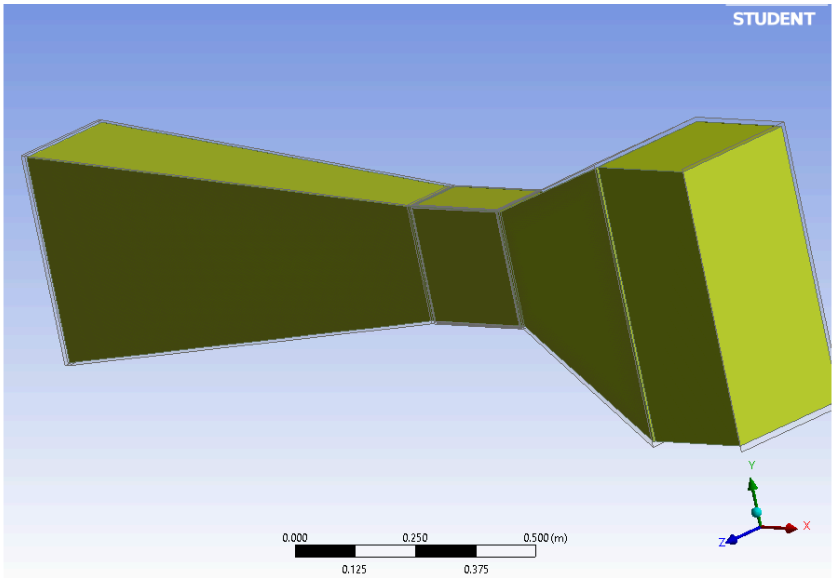

Figure 1-12. Inside Fluid Highlighted

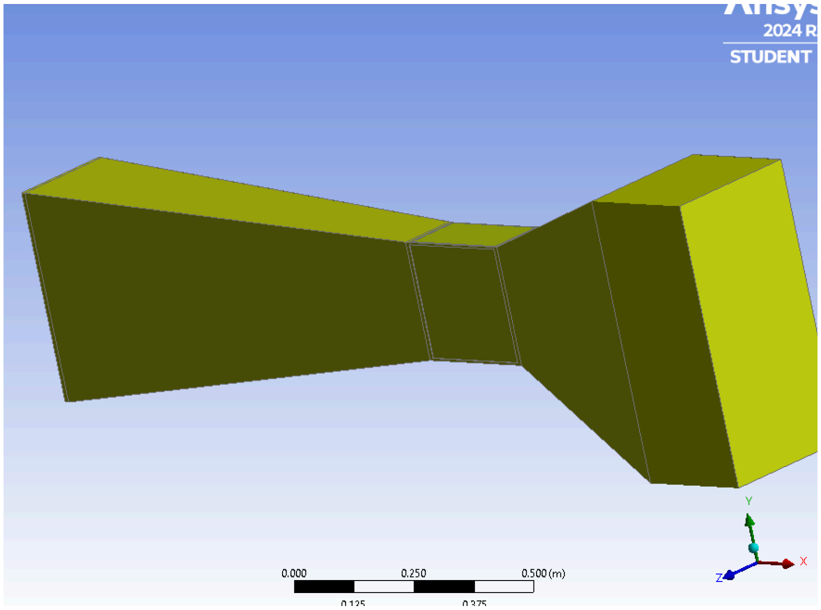

Figure 1-13. Shell Highlighted

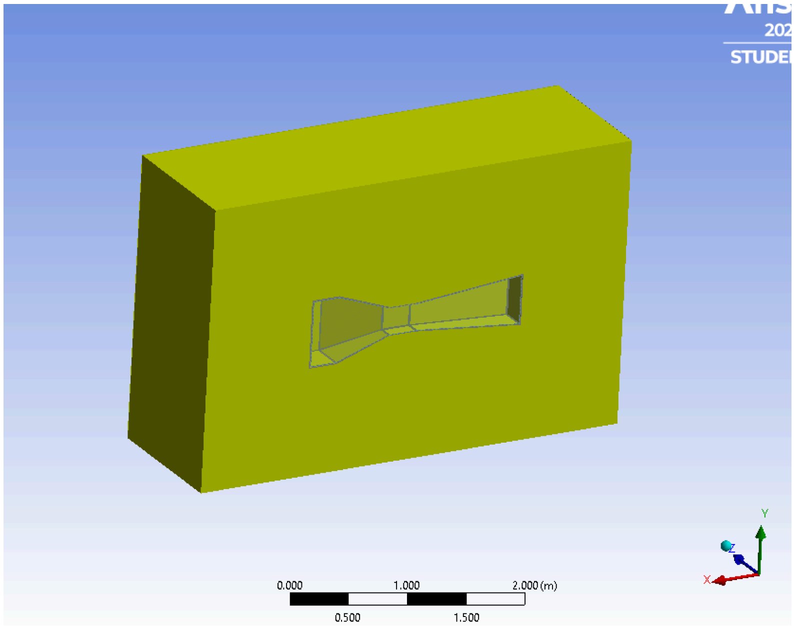

Figure 1-14. Solid Highlighted

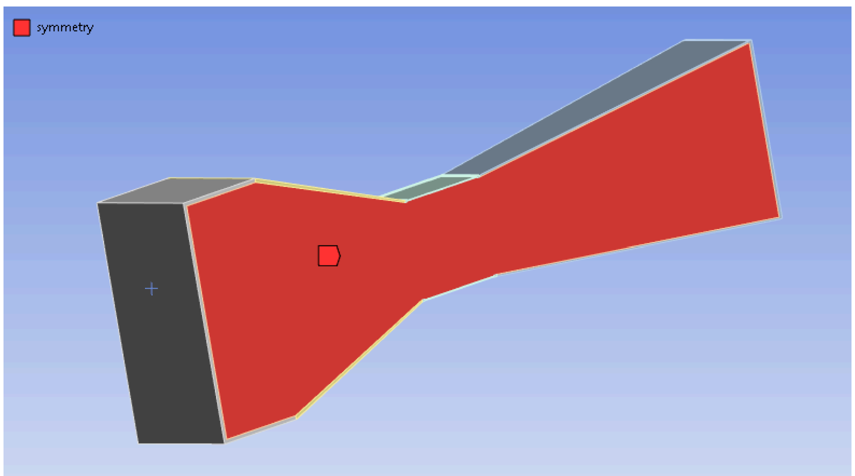

Figure 1-15. Named Selection Example

5. Conclusion

Diffuser

As the angle of attack decreased from ±12.95° above parallel, to ±11.31° above parallel, to ±10.20° above parallel, the recirculation zone following the test area decreased drastically. This decrease in recirculation of airflow will result in higher accuracy when conducting experiments within the wind tunnel. Based on these findings, the angle of attack of the diffuser portion of the wind tunnel plays a major factor in recirculation zones. We believe the most effective diffuser should be ±10° above parallel to create a wind tunnel that produces consistent results. Depending on the figure being tested within the tunnel and the airflow velocity, results may vary.

Uniform Flow in Test Section

At each line (test section entrance, middle, and end) the velocity magnitude plot showed similar results. Each plot shows a constant velocity magnitude across the line (top wall to bottom wall). Figures 1-7 through 1-10 show the resulting velocity magnitude to be 110 m/s. Two geometries were also compared to see if the uniformity of the flow varies with the change of the angle of the diffuser. By comparing Figures 1-8 and 2-4, the change of angle did not affect the uniformity of the flow as the results remained the same. Along with these results, modifying the geometry by implementing filets to reduce the amount of sharp edges seemed promising to improve the uniformity of the flow inside the wind tunnel. However, after comparing the results of the original and modified 36 x 36 model as seen in Figure 2-8, the effects were mostly negligible. The next step to testing for uniform flow in the test section would be to research alternative effects (such as the flow straightener we wanted to test originally) to the geometry that would cause more of a change to the flow quality. A challenge we ran into was how to set a flow to go through a flow straightener in this software. In the alternative method section (Figures 1-11 through 1-15) this was experimented, although we did switch back to a solid geometry to continue and finish the project by the deadline.

The fine mesh velocity magnitude plot with a cell count of close to 1 million provided similar results to our course mesh. This shows our use of the more coarse mesh for the majority of this project is adequate to provide accurate results.

Tangential Inlet Model

Despite a large majority of subsonic wind tunnels implementing a tangential inlet design for their inlet, our results show little to meaningful increase in producing uniform flow through the test section. When comparing Figures 1-8, 2-4 and 4-9, the change of inlet geometry did not affect the uniformity of the flow as the results remained the same. We believe that further improvements such as increasing the size of the inlet surface area could produce a meaningful increase to uniformity in the Test Section.

A notable difference is that the Tangential Inlet Model produced areas of little to no circulation within the diffuser, see Figures 4-5 and 4-7. Further testing may find that a configuration of the Tangential Inlet and a cylindrical diffuser geometry could reduce recirculation zones to near negligible levels.

6. References

[1] Barlow, Jewel B., William H. Rae, and Alan Pope. Low-speed wind tunnel testing. John wiley & sons, 1999.

[2] A.S. Bahaj, A.F. Molland, J.R. Chaplin, W.M.J. Batten, Power and thrust measurements of marine current turbines under various hydrodynamic flow conditions in a cavitation tunnel and a towing tank, Renewable Energy, Volume 32, Issue 3, 2007.

[3] Cattafesta, Louis, Chris Bahr, and Jose Mathew. Fundamentals of wind-tunnel design. Encyclopedia of aerospace engineering, 2010.

7. Appendix

Contributions of Each Team Member

In general Figures numbering will be used to delineate which plots, and in-turn tests, were completed by each member. Member number ties to the first digit of the Figure identifier number; example "Figure 3-10" denotes the tenth figure that member number 3 produced. This numbering system results in easier identification of roles and allows for each member to keep track of their own figures within this report. Figures 0-1, 2,3, etc. will denote figures that show settings that are present for all tests completed by each member.

<u>Zachery Taylor - Member #3</u>

Responsible for creating the scaled base geometry with a 46x46mm diffuser outlet, the 40x40mm diffuser variant, and the 36x36mm diffuser variant. Completing all initial mesh independency verifications to confirm that a 1.6mm mesh will produce sufficient and accurate results for future testing. Testing Velocity Vectors and Velocity Pathlines for the entire wind tunnel as well as areas near wall planes. This testing was completed to determine recirculation zones and experiment with the different diffuser geometries to reduce such zones. Completed all report sections regarding the diffuser and recirculations zones, as well as detailed description of the base geometry. Completed thorough Abstract portion of the report as well as an insightful Introduction to describe the tests that were performed.

<u>Juliana Barchie - Member #1</u>

Responsible for creating an in-depth proposal that was submitted for feedback from the professor and adjusting the scope of the project based on the feedback. Completing tests on the original model which included, but not limited to: Creating and testing a 1.16mm mesh with near 1,000,000 cells - as recommended by professor; Reynold's number calculation and testing to confirm all models experienced similar environments; Determination of Solution Method; Residual Testing on Original Model (46x46mm outlet); XY Velocity Plots at 3 different locations within the Test Section on the Original Model. Member also effectively used her time to test an alternative modeling method that was not taught in the course. Displayed the team's knowledge of reasoning while using this software: convergence, mesh independence, utilizing details from the three projects for professional display (curve display characteristics), use of Re for flow settings, etc.

<u>Terry Malinowski - Member #2</u>

Responsible for Residual Testing on Optimized Model (36x36mm outlet); XY Velocity Plots at 3 different locations within the Test Section on the Optimized Model. Member also responsible for creating a separate geometry with rounded (filet) wall edges. This improved model was used to run XY velocity plots used for comparison to the modified 36x36mm model along with an XY plot comparing the midpoint of each of the two models to compare uniformity. Main goal was testing to determine if smoothing the edges will increase uniformity throughout the test section and overall improve the design of the wind tunnel. Discussion regarding stream uniformity and the results of these performed tests.

<u>D'Andre Walden - Member #4</u>

Responsible for creating geometry with modified inlet that is tangential to the Test Section. This member also completed all baseline tests on this geometry to give a holistic view of how this modification affected the flow. Tests included, but not limited to: Creating and testing a 1.6mm mesh on the Tangential Inlet Geometry; Residual Testing on Tangential Inlet Model (46x46mm outlet); XY Velocity Plots at 3 different locations within the Test Section on the Tangential Inlet Model; Static Pressure Contour Plot along symmetry of Tangential Inlet Model. Member also produced Velocity Vectors and Velocity Pathlines plots for the entire wind tunnel as well as areas near wall planes on the Tangential Inlet Model.